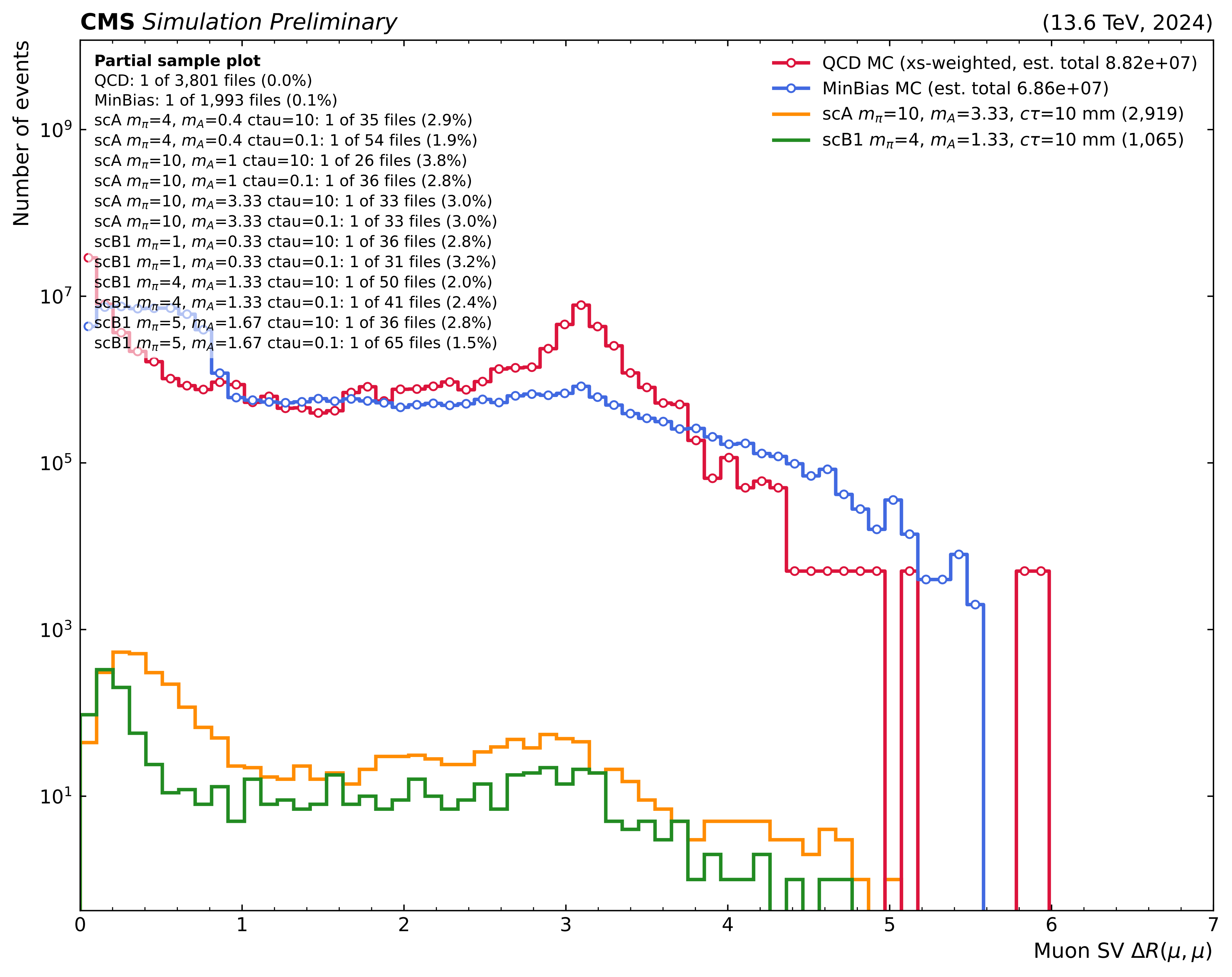
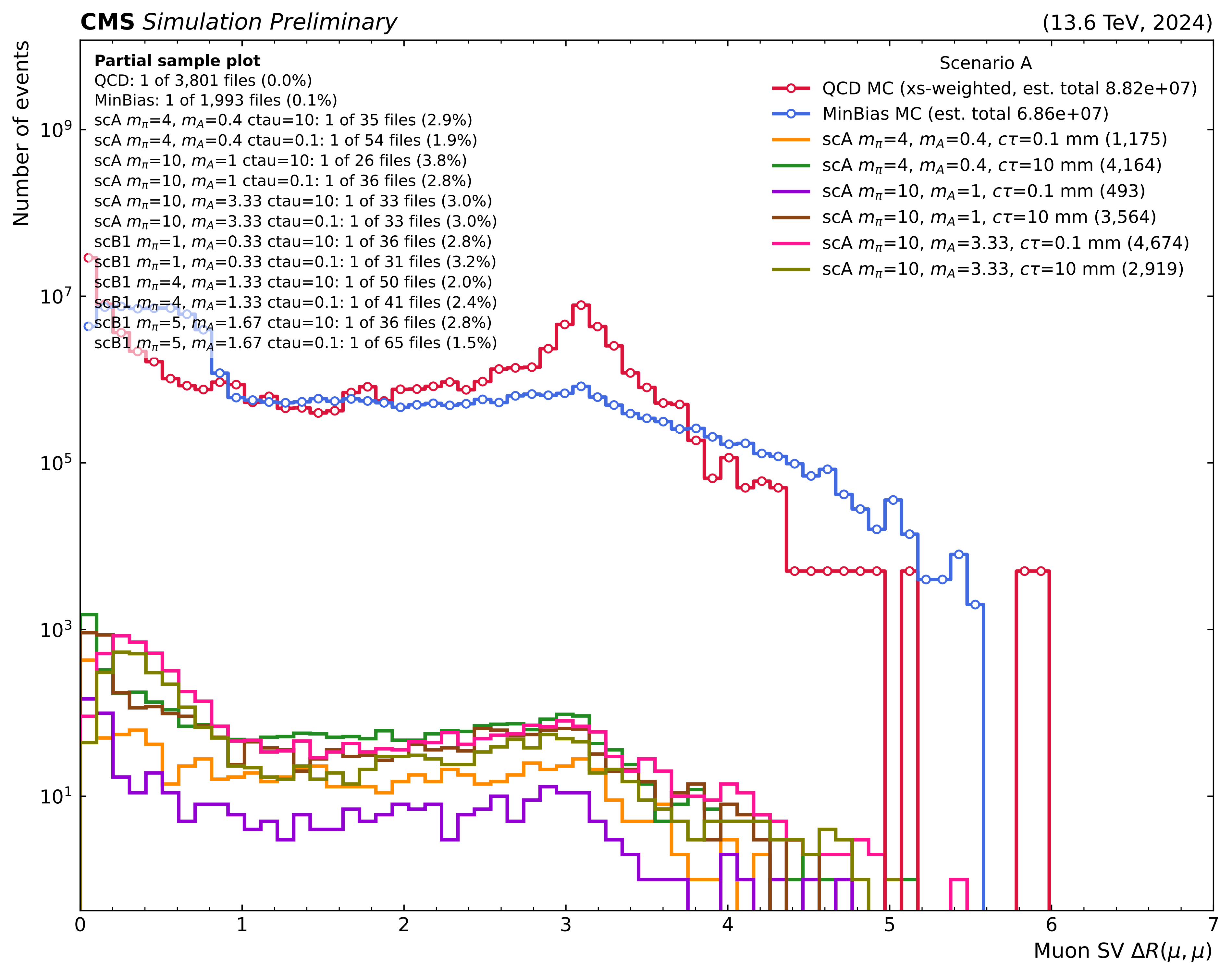
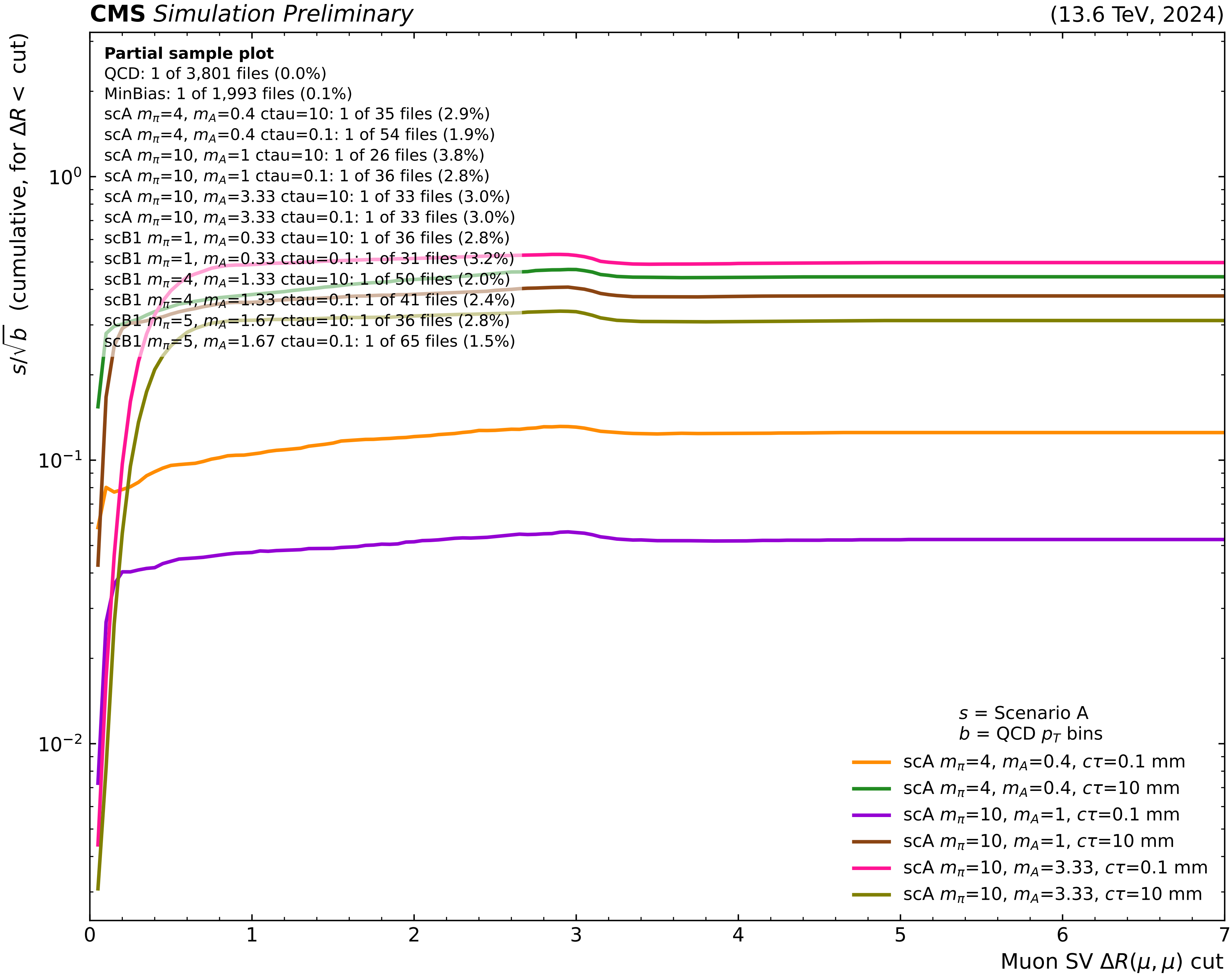
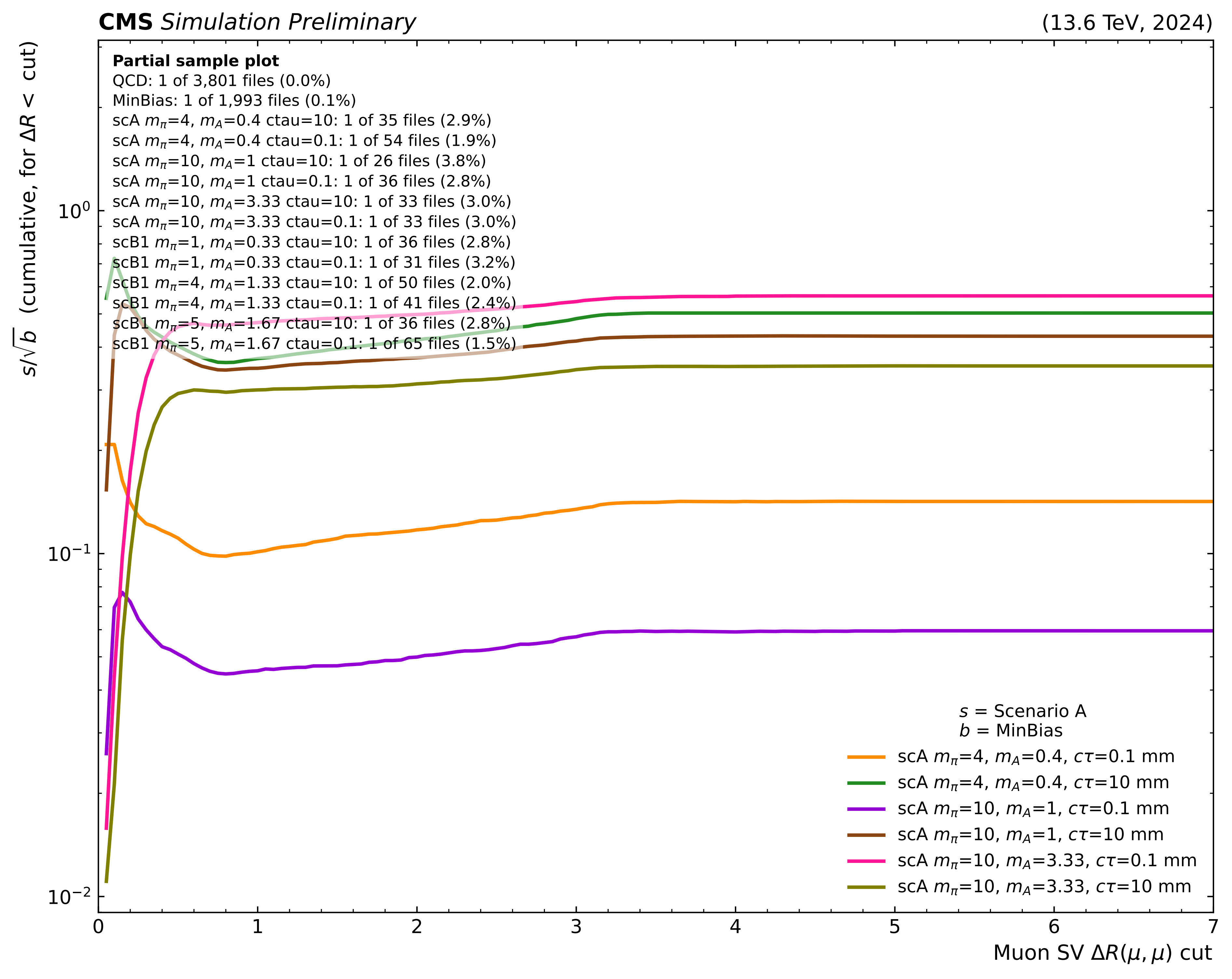

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
Number of events
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$

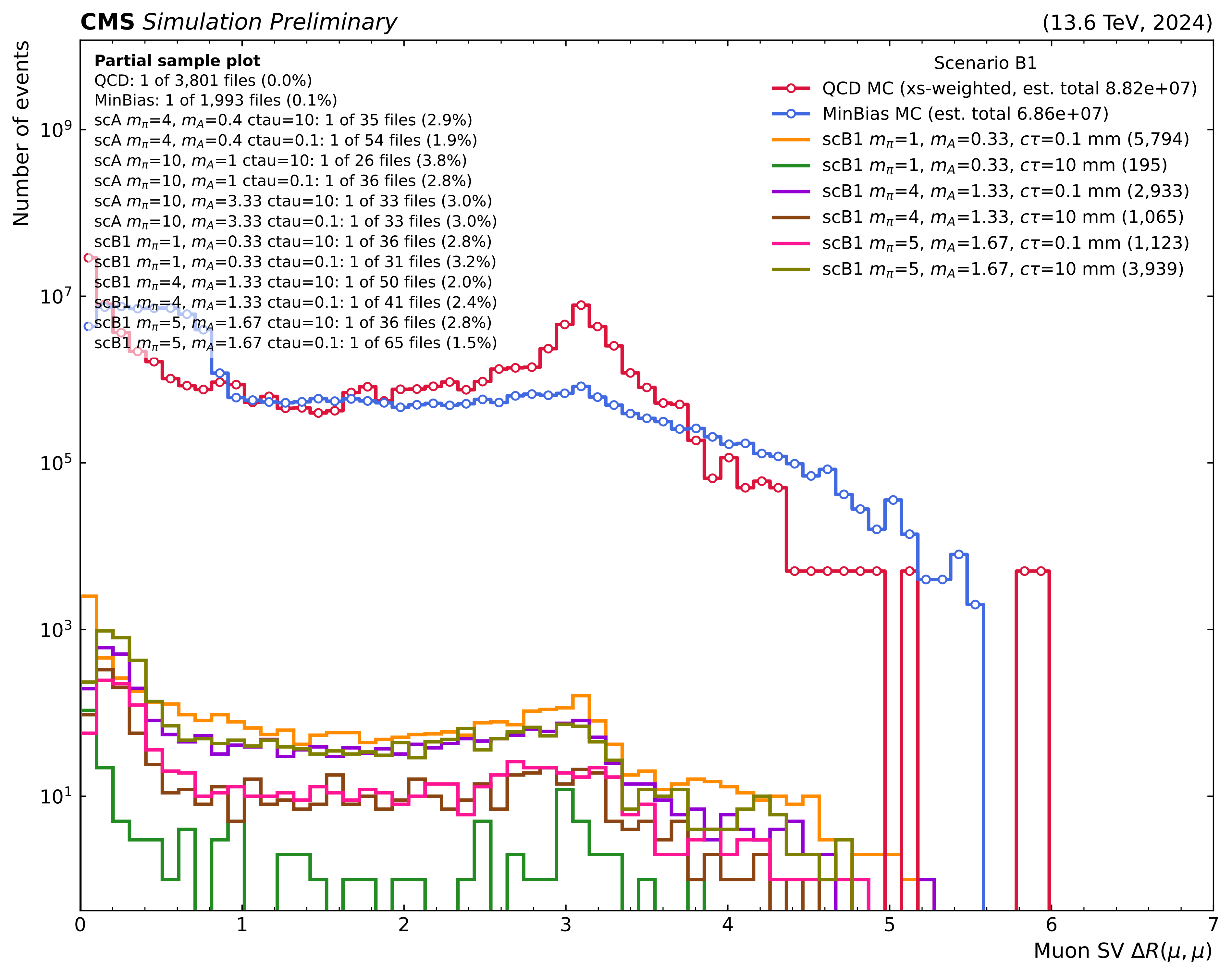


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario A
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm (1,175)
scA m_π=4, m_A=0.4, $c\tau$=10 mm (4,164)
scA m_π=10, m_A=1, $c\tau$=0.1 mm (493)
scA m_π=10, m_A=1, $c\tau$=10 mm (3,564)
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm (4,674)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
Number of events
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$

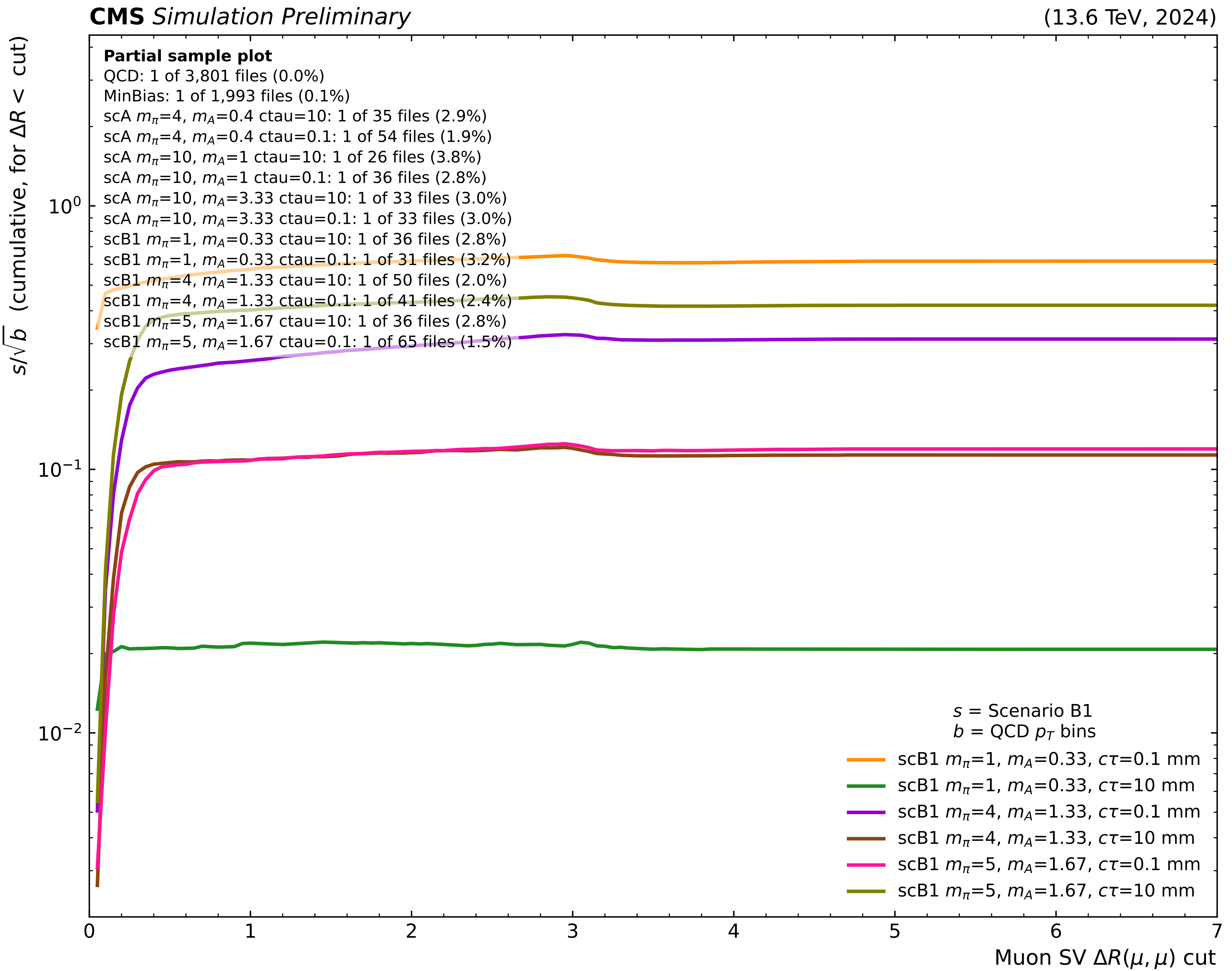

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for ΔR < cut)
10^0
10^{-1}
10^{-2}
s = Scenario A
b = QCD p_T bins
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$ cut

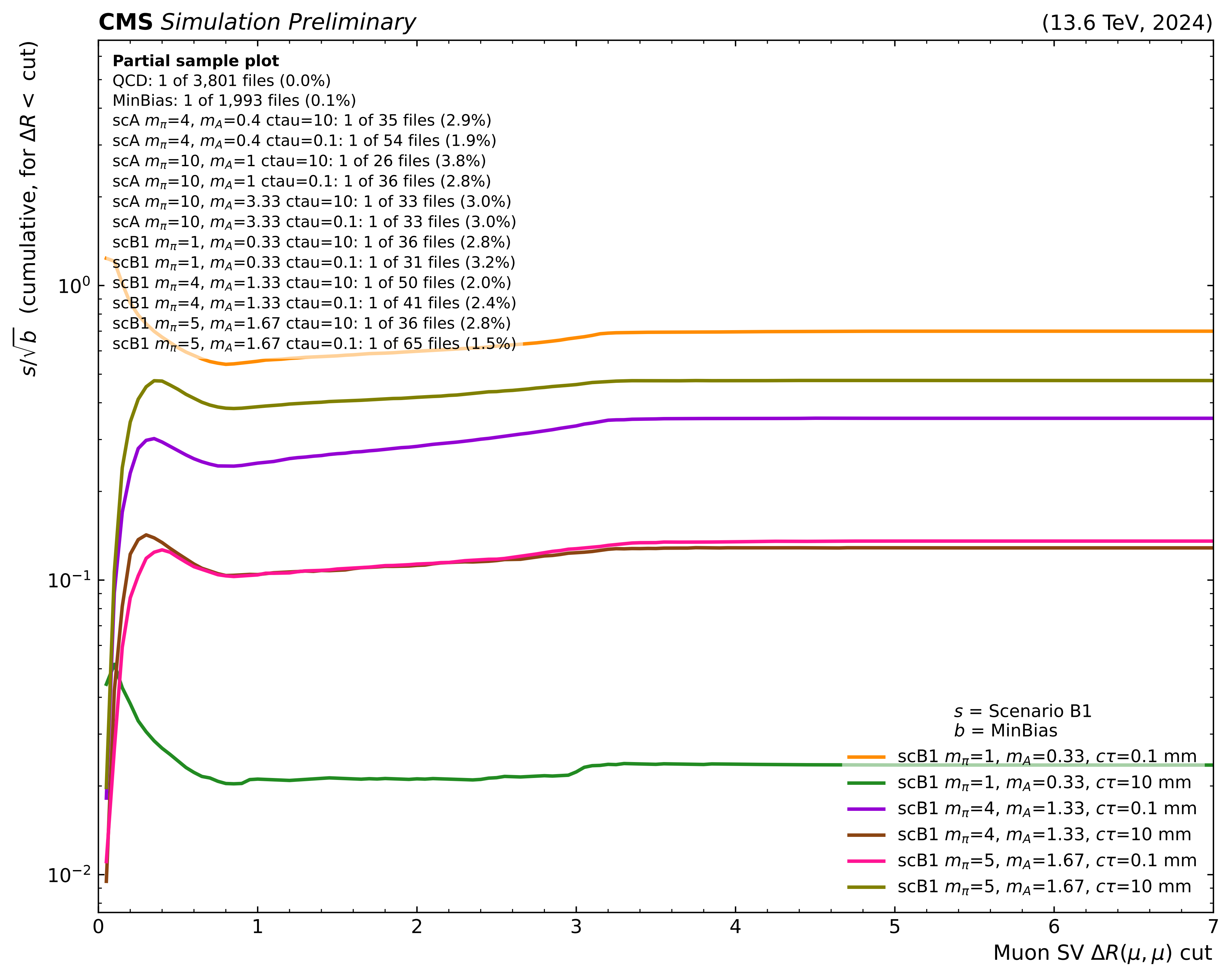


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for ΔR < cut)
10^0
10^{-1}
10^{-2}
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$ cut
s = Scenario A
b = MinBias
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm

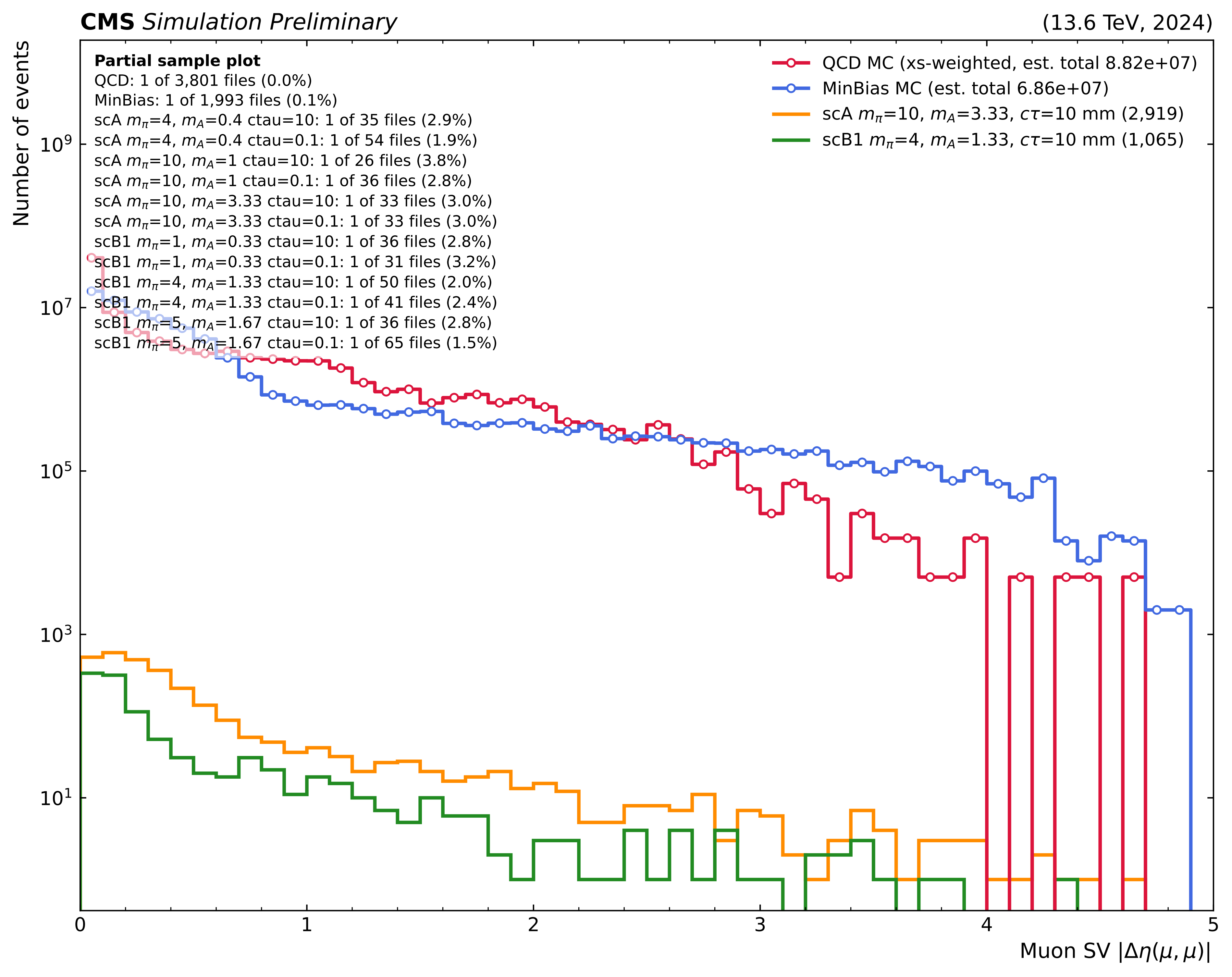


CMS Simulation Preliminary
(13.6 TeV, 2024)
Number of events
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario B1
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm (5,794)
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm (195)
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm (2,933)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm (1,123)
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm (3,939)
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$

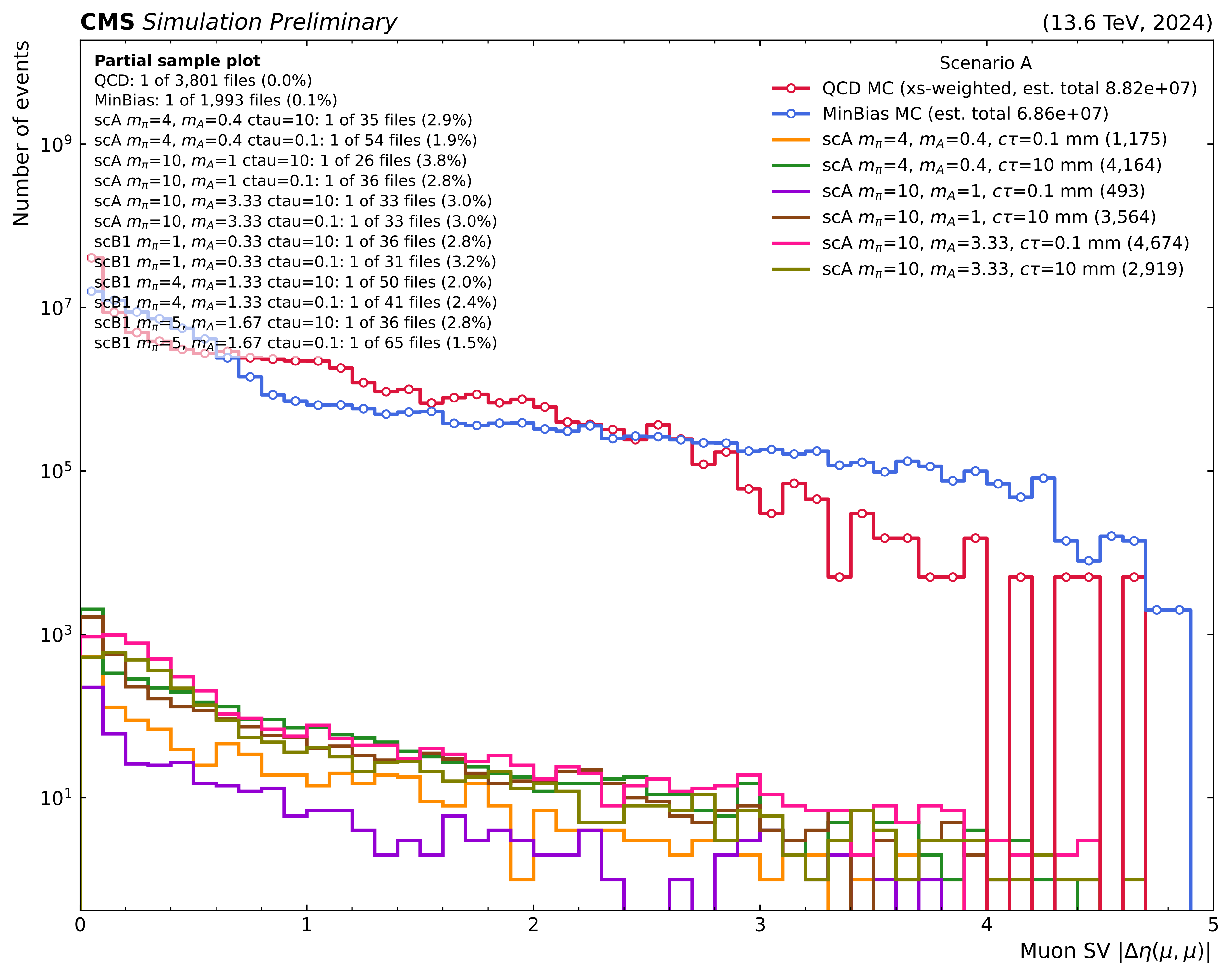

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for ΔR < cut)
10^0
10^{-1}
10^{-2}
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$ cut
s = Scenario B1
b = QCD p_T bins
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm



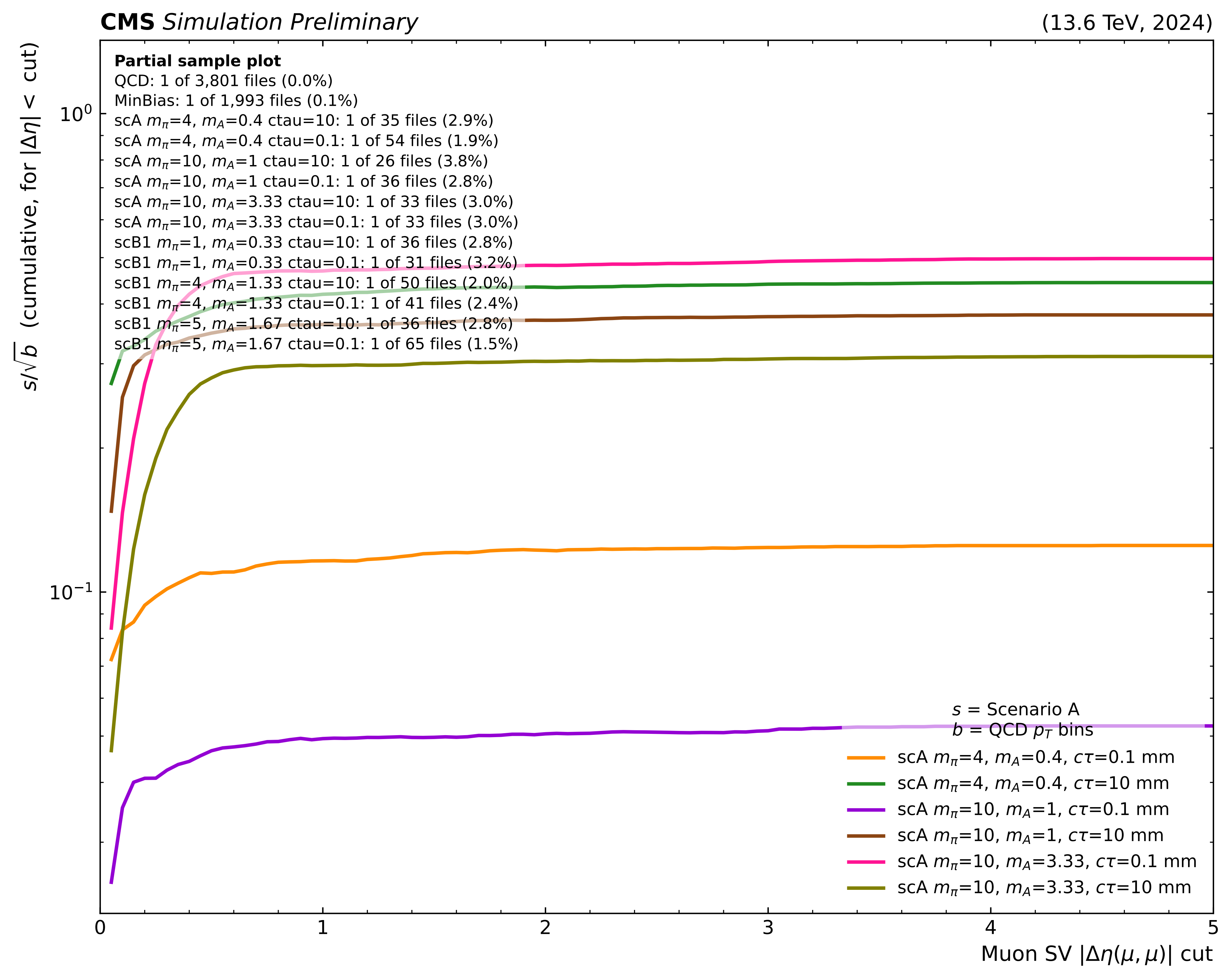
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $\Delta R <$ cut)
10^0
10^{-1}
10^{-2}
s = Scenario B1
b = MinBias
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0
1
2
3
4
5
6
7
Muon SV $\Delta R(\mu,\mu)$ cut



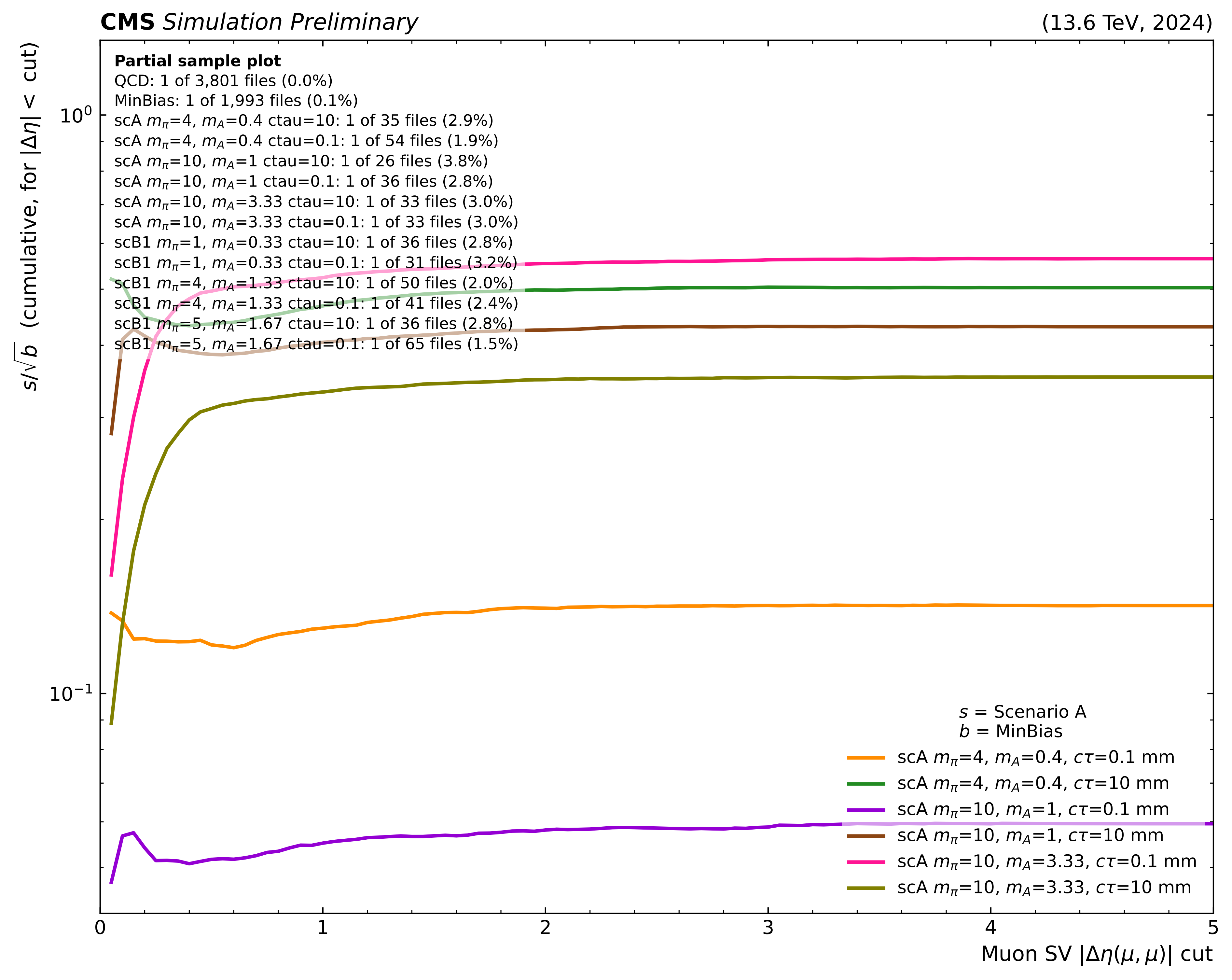
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
Number of events
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$

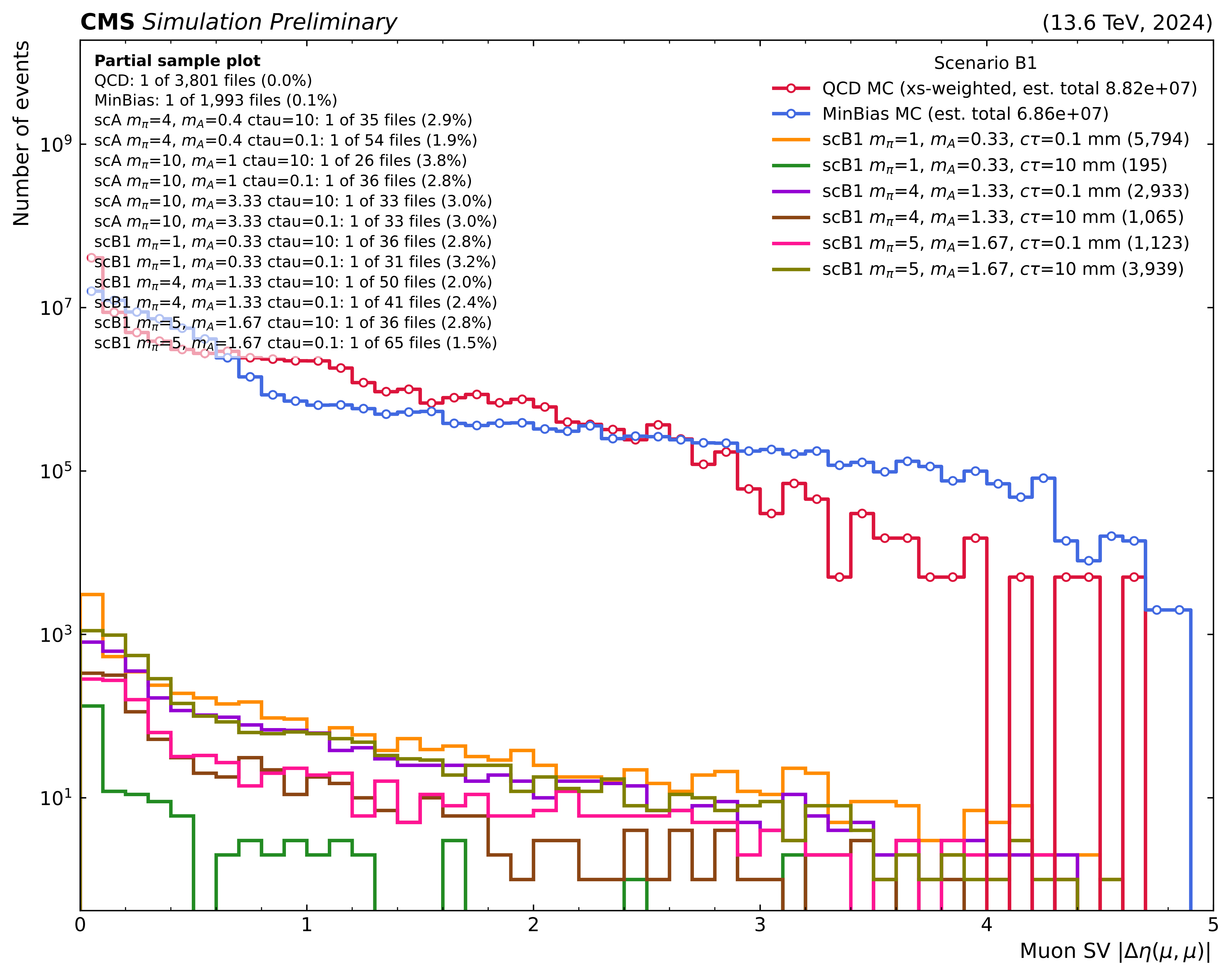


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario A
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm (1,175)
scA m_π=4, m_A=0.4, $c\tau$=10 mm (4,164)
scA m_π=10, m_A=1, $c\tau$=0.1 mm (493)
scA m_π=10, m_A=1, $c\tau$=10 mm (3,564)
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm (4,674)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
Number of events
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$

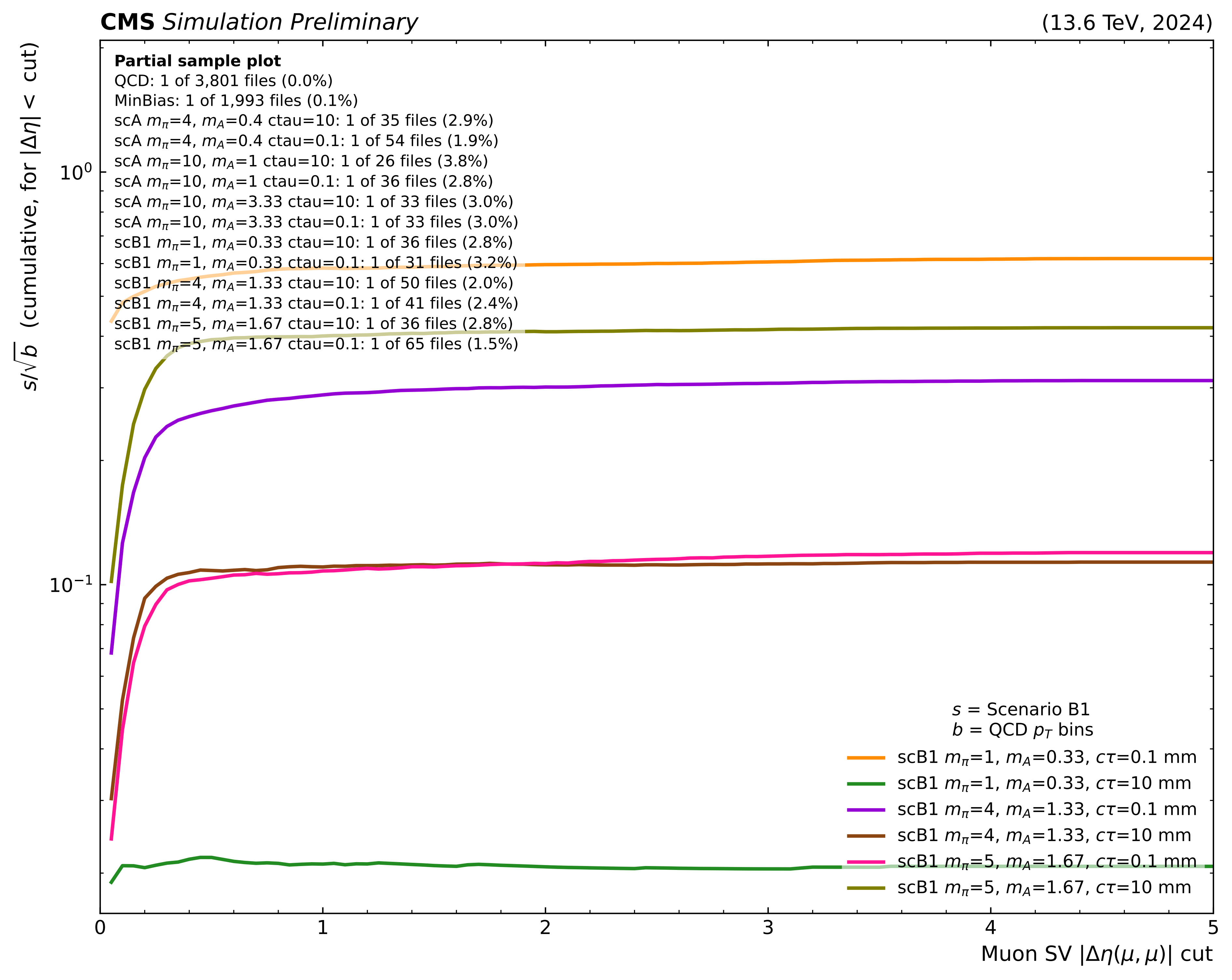


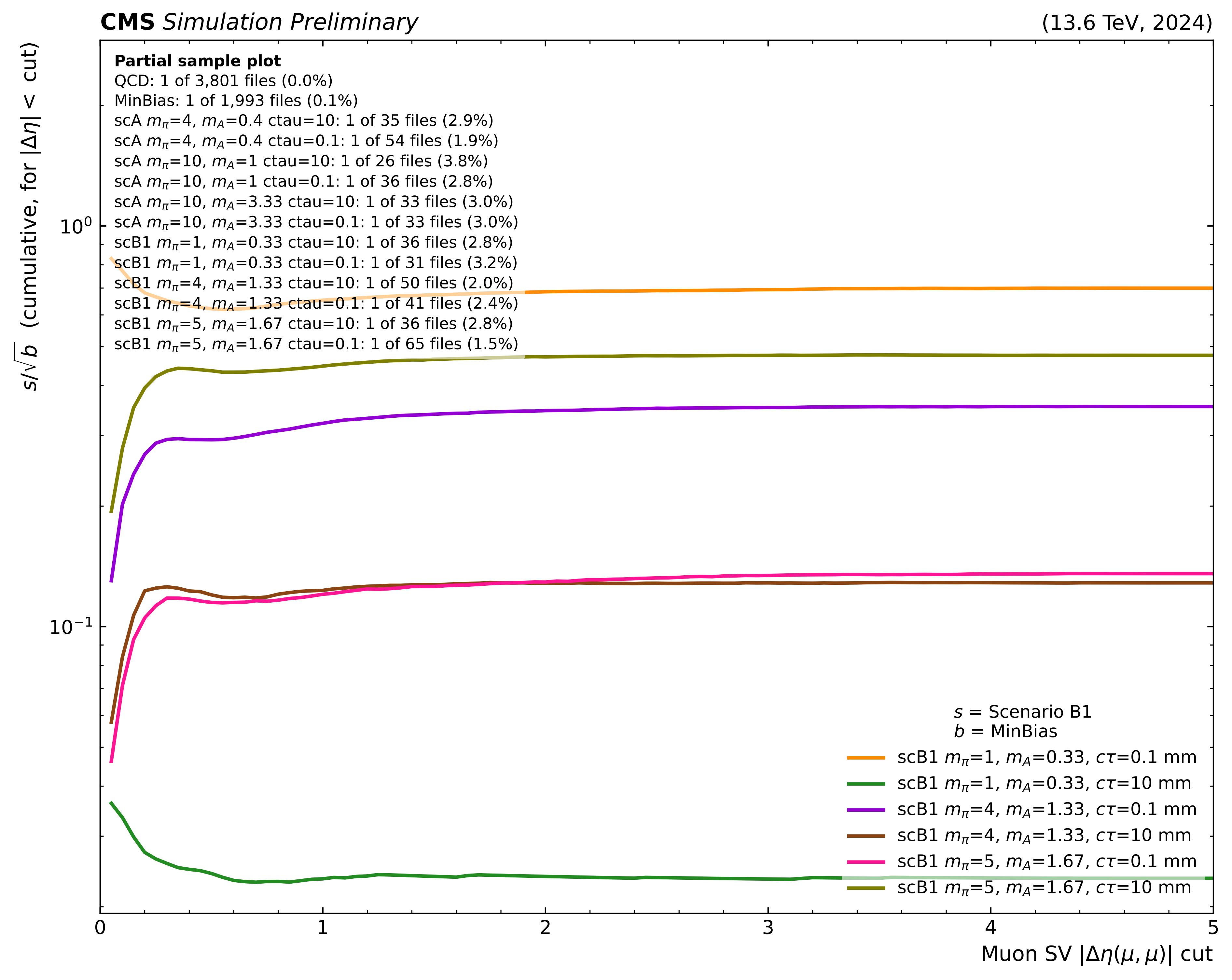
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\eta|$ < cut)
10^0
10^{-1}
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$ cut
s = Scenario A
b = QCD p_T bins
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\eta|$ < cut)
10^0
10^{-1}
s = Scenario A
b = MinBias
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$ cut

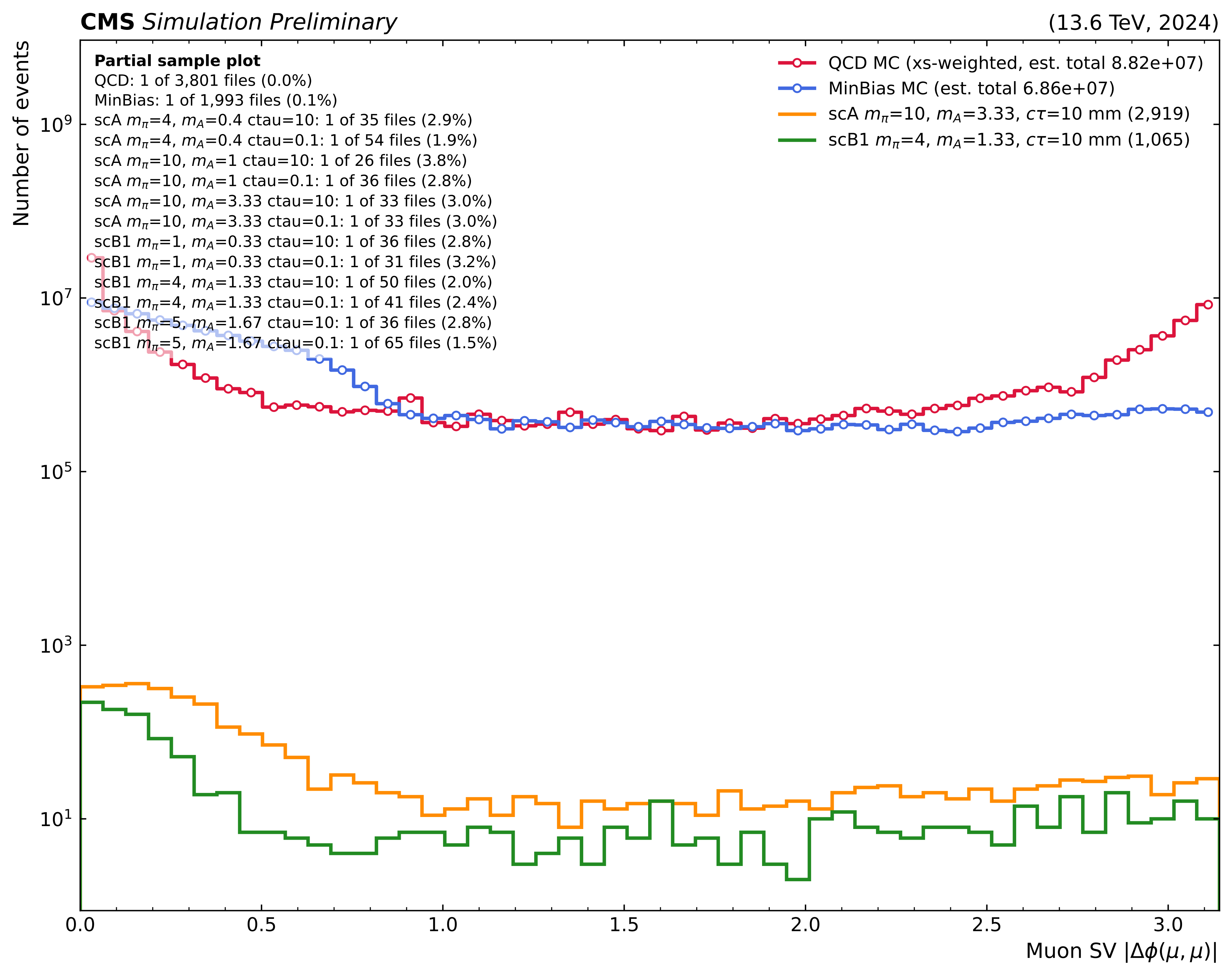


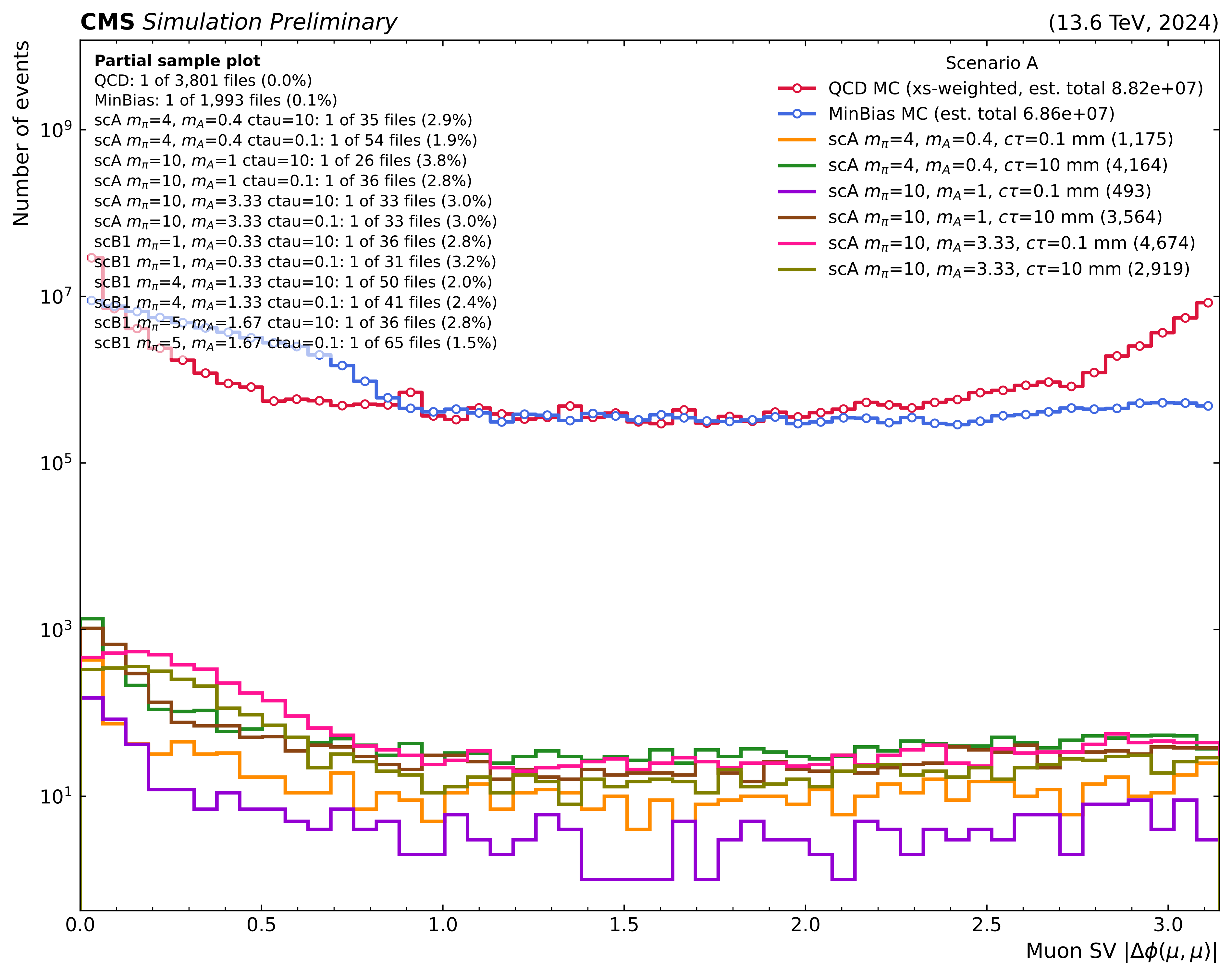
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario B1
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm (5,794)
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm (195)
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm (2,933)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm (1,123)
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm (3,939)
Number of events
10^9
10^7
10^5
10^3
10^1
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$

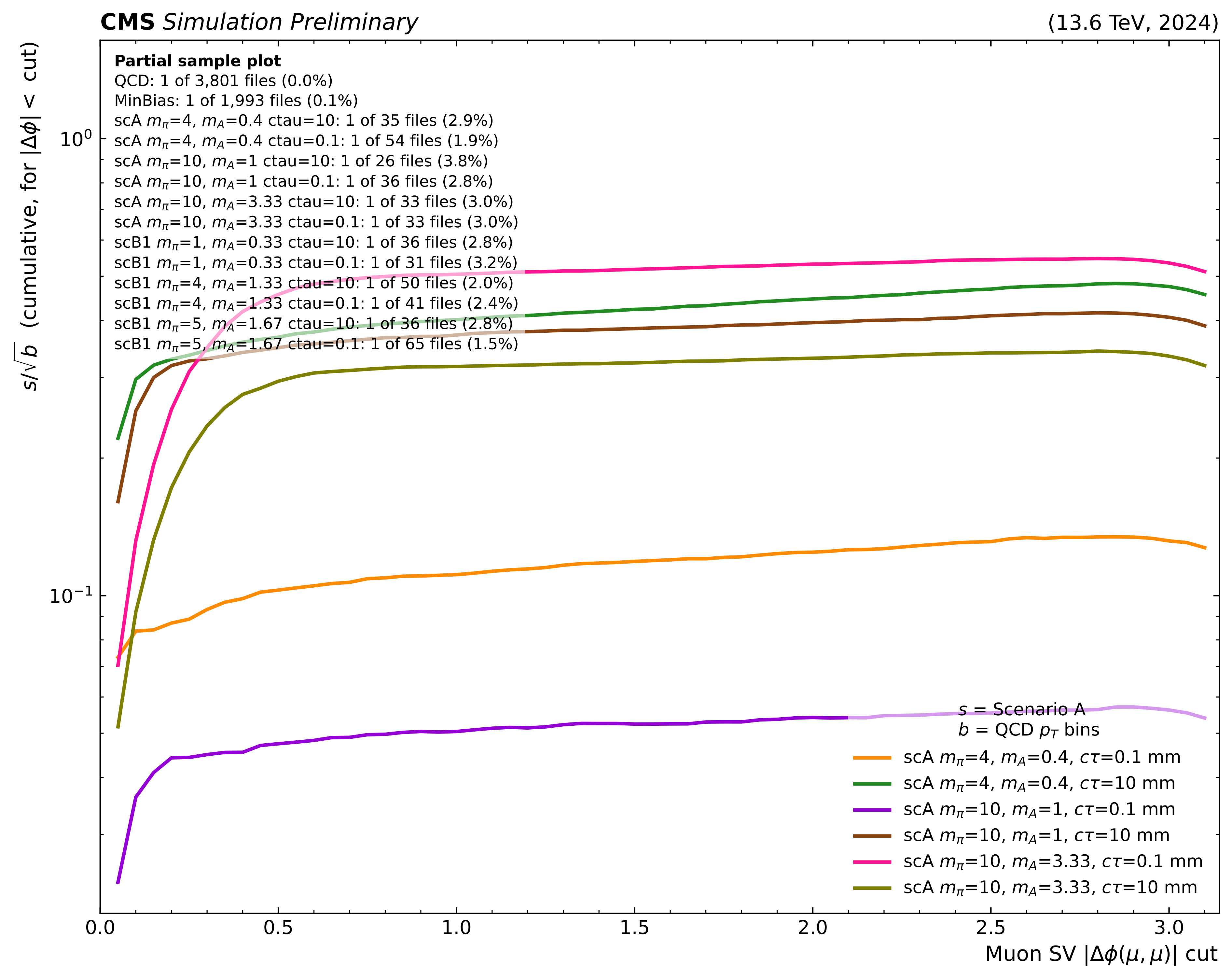

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\eta|$ < cut)
10^0
10^{-1}
s = Scenario B1
b = QCD p_T bins
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$ cut

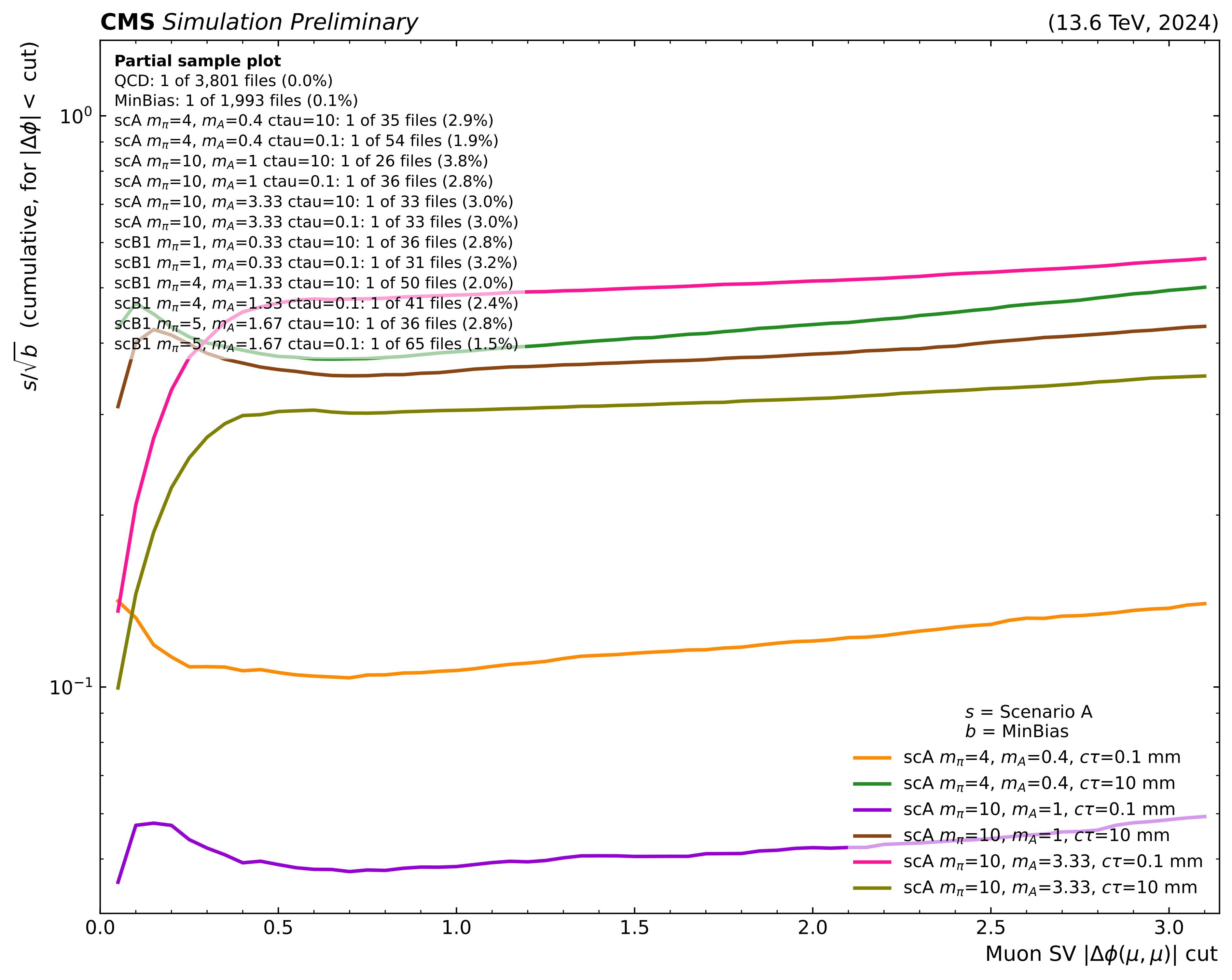


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\eta|$ < cut)
10^0
10^{-1}
s = Scenario B1
b = MinBias
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0
1
2
3
4
5
Muon SV $|\Delta\eta(\mu,\mu)|$ cut

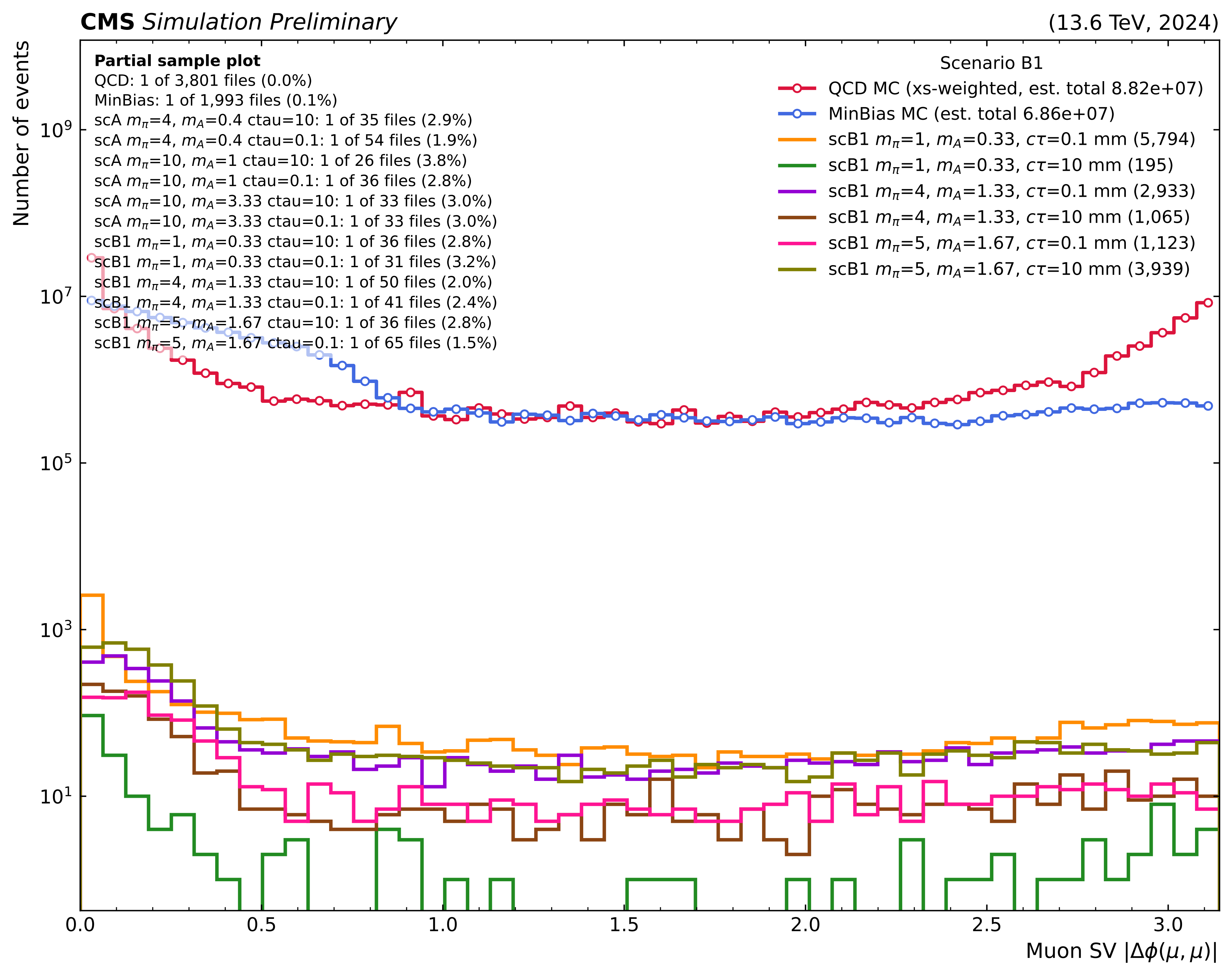

CMS Simulation Preliminary
(13.6 TeV, 2024)
Number of events
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$

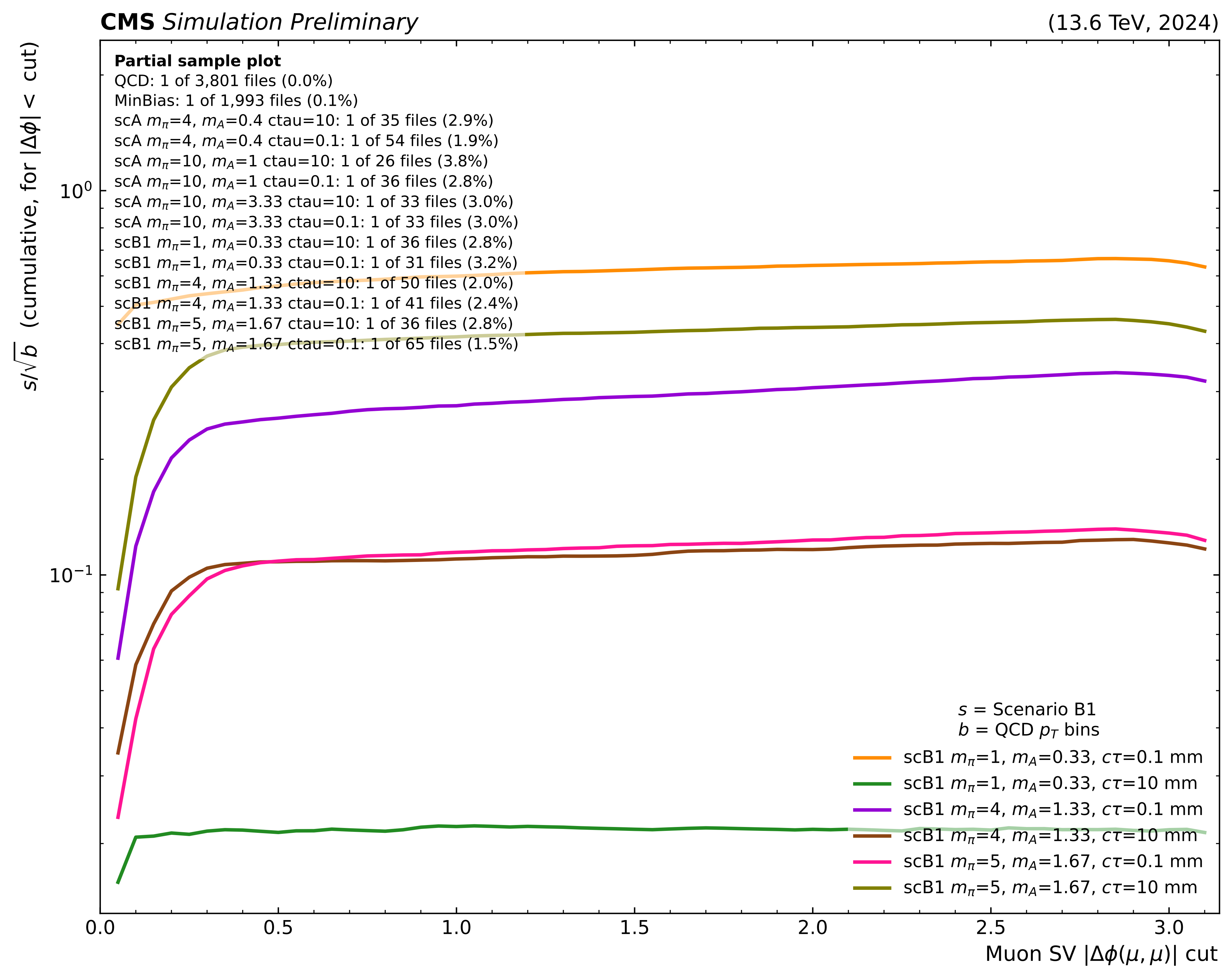

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario A
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm (1,175)
scA m_π=4, m_A=0.4, $c\tau$=10 mm (4,164)
scA m_π=10, m_A=1, $c\tau$=0.1 mm (493)
scA m_π=10, m_A=1, $c\tau$=10 mm (3,564)
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm (4,674)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
Number of events
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\phi|$ < cut)
10^0
10^{-1}
s = Scenario A
b = QCD p_T bins
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$ cut

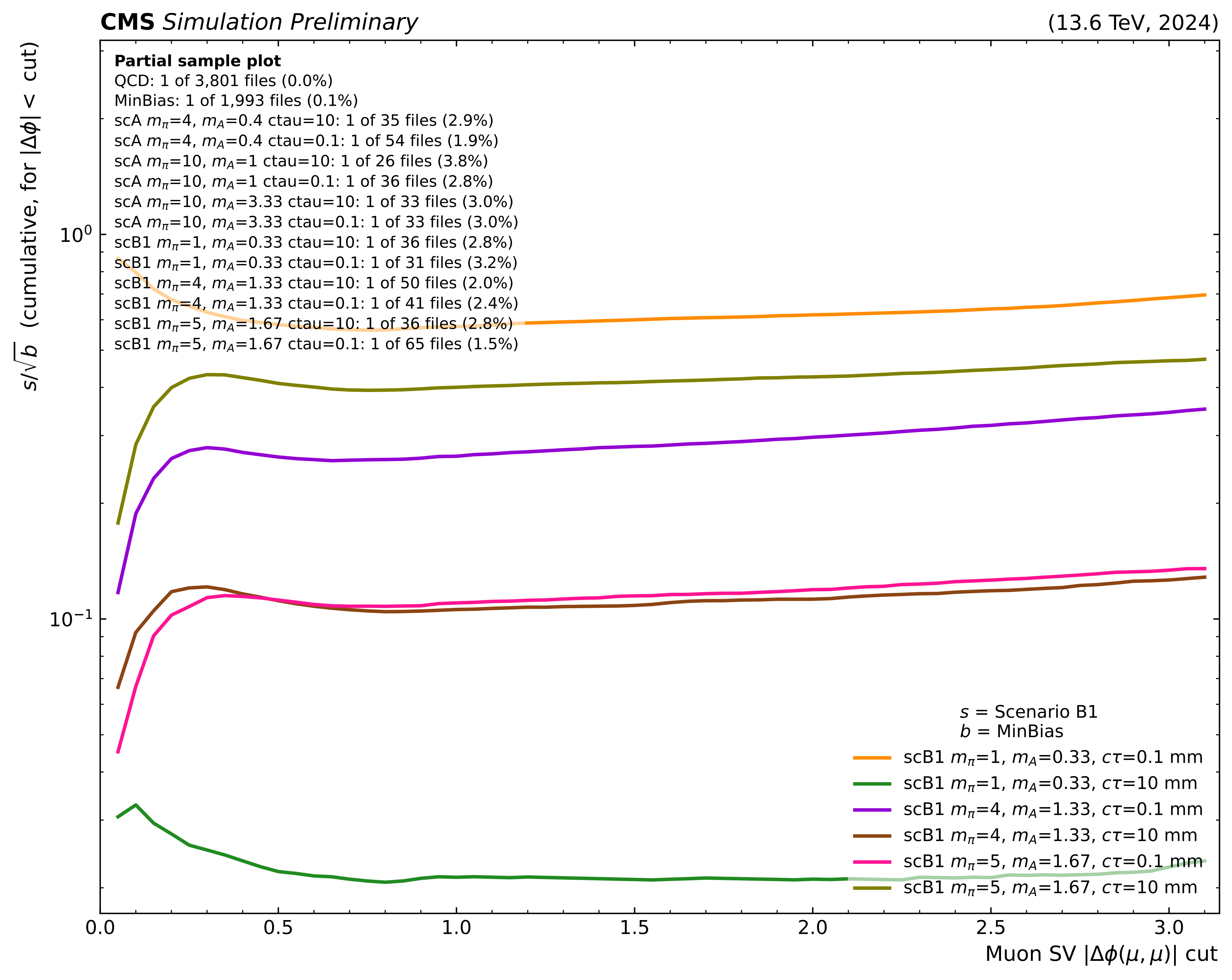


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\phi|$ < cut)
10^0
10^{-1}
s = Scenario A
b = MinBias
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$ cut

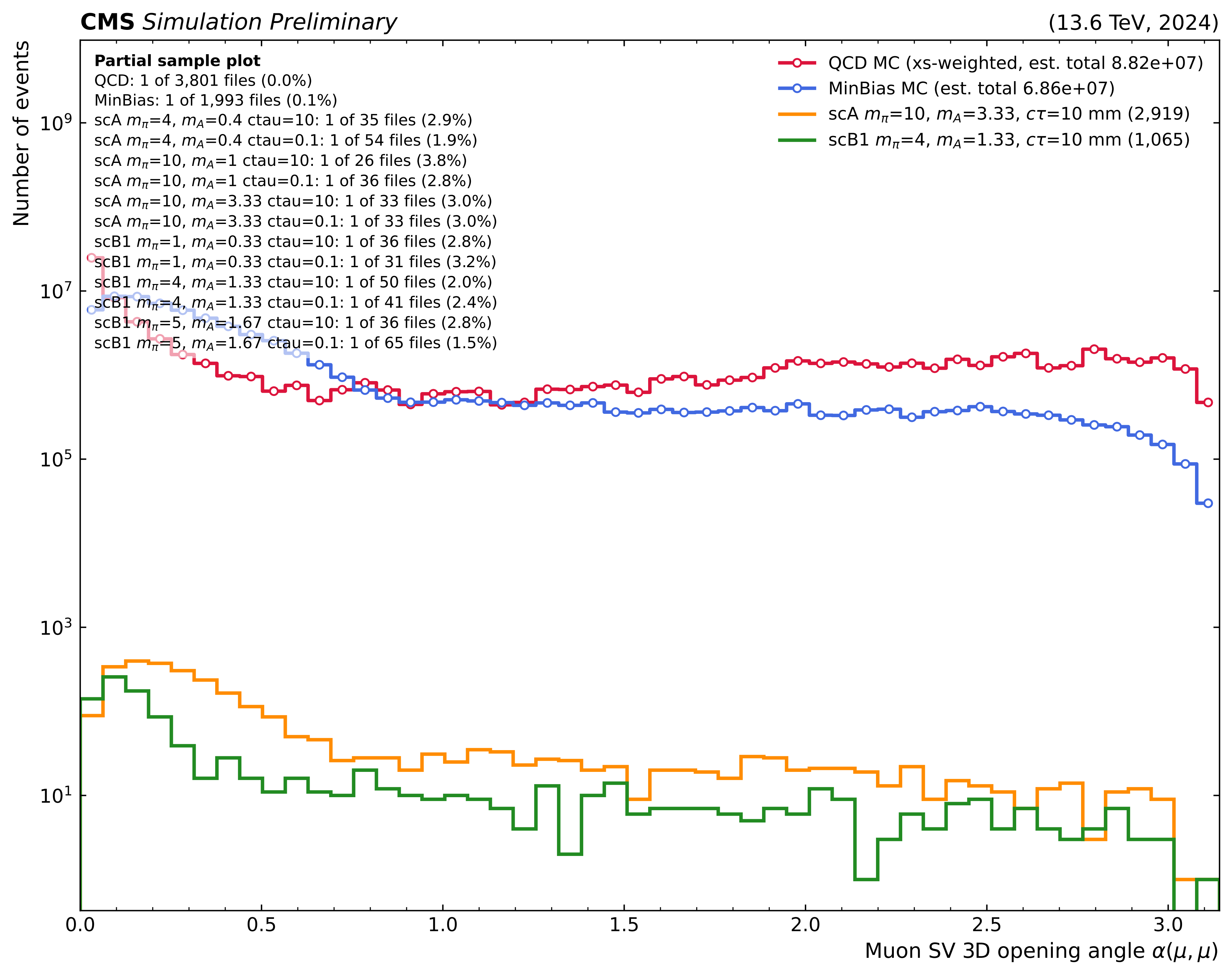

CMS Simulation Preliminary
(13.6 TeV, 2024)
Number of events
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario B1
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm (5,794)
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm (195)
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm (2,933)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm (1,123)
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm (3,939)
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$



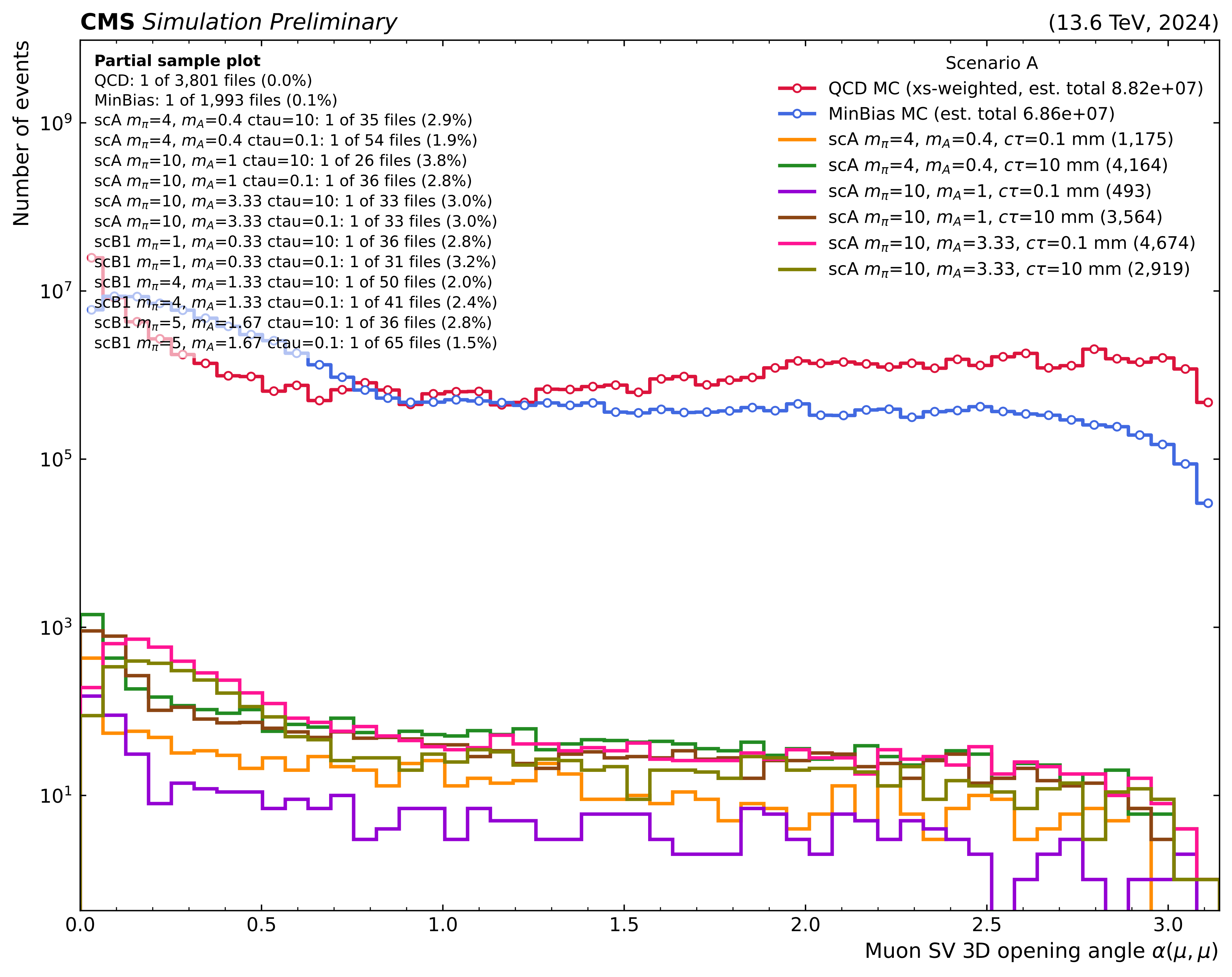
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\phi|$ < cut)
10^0
10^{-1}
s = Scenario B1
b = QCD p_T bins
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$ cut

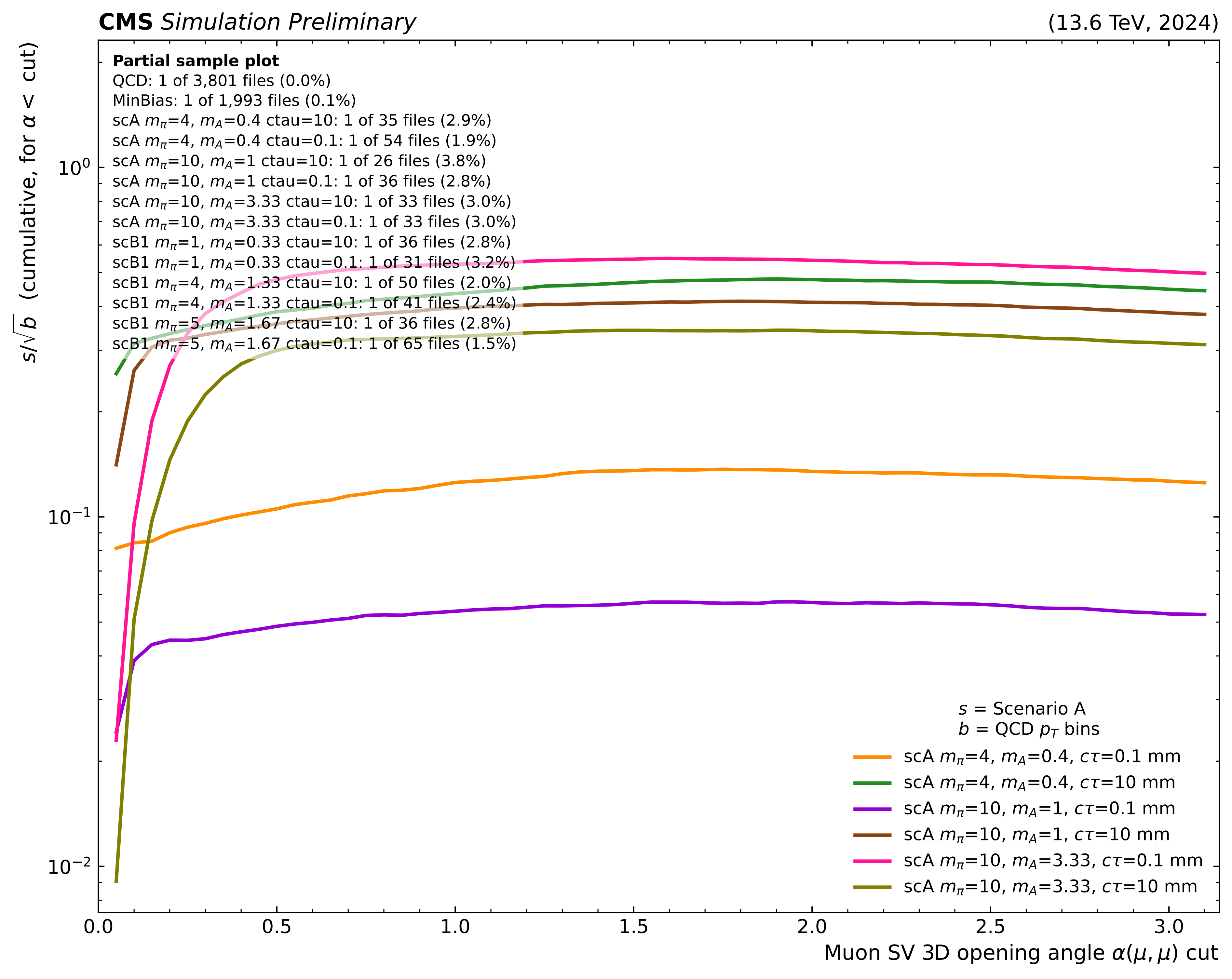


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $|\Delta\phi|$ < cut)
10^0
10^{-1}
s = Scenario B1
b = MinBias
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV $|\Delta\phi(\mu,\mu)|$ cut

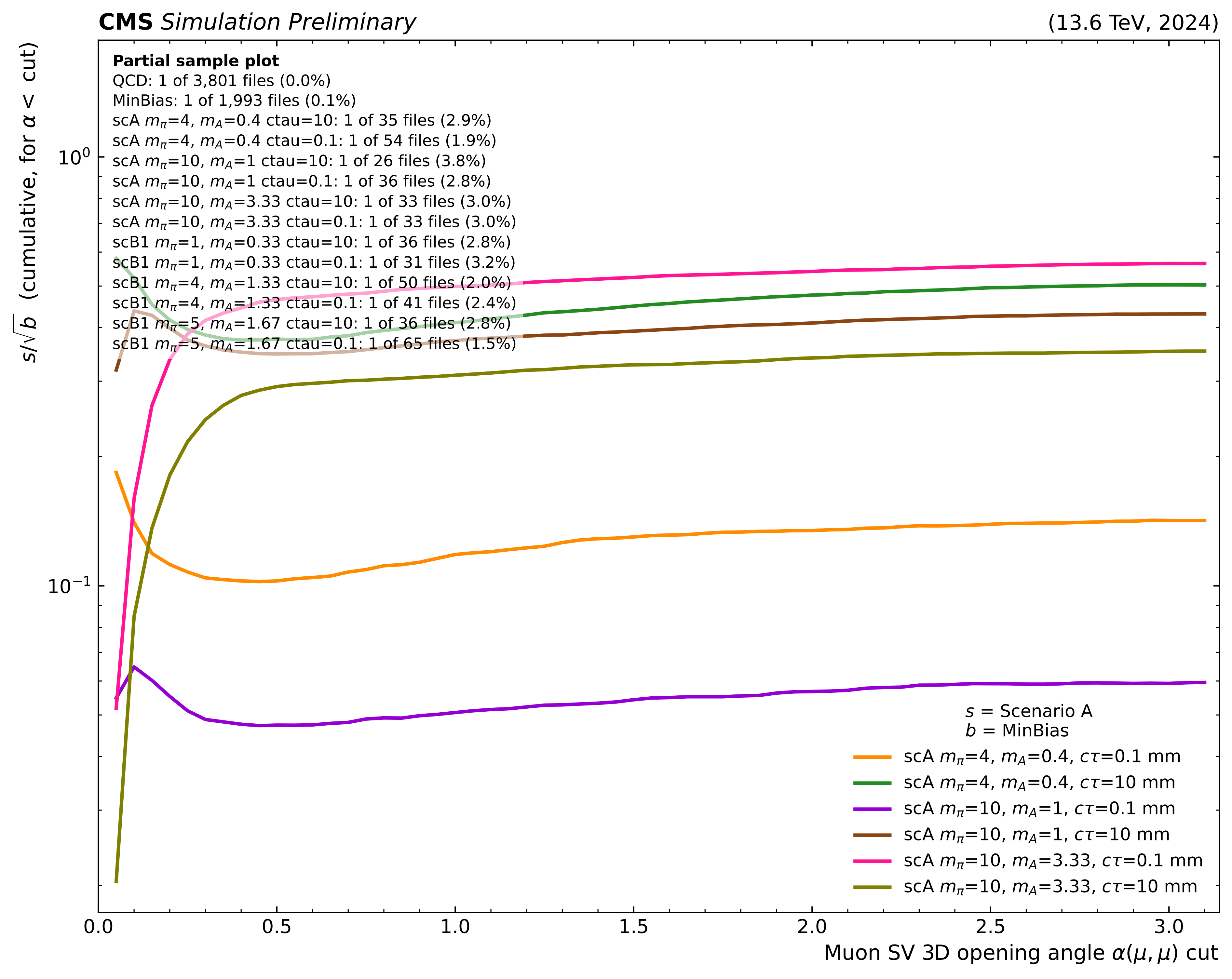


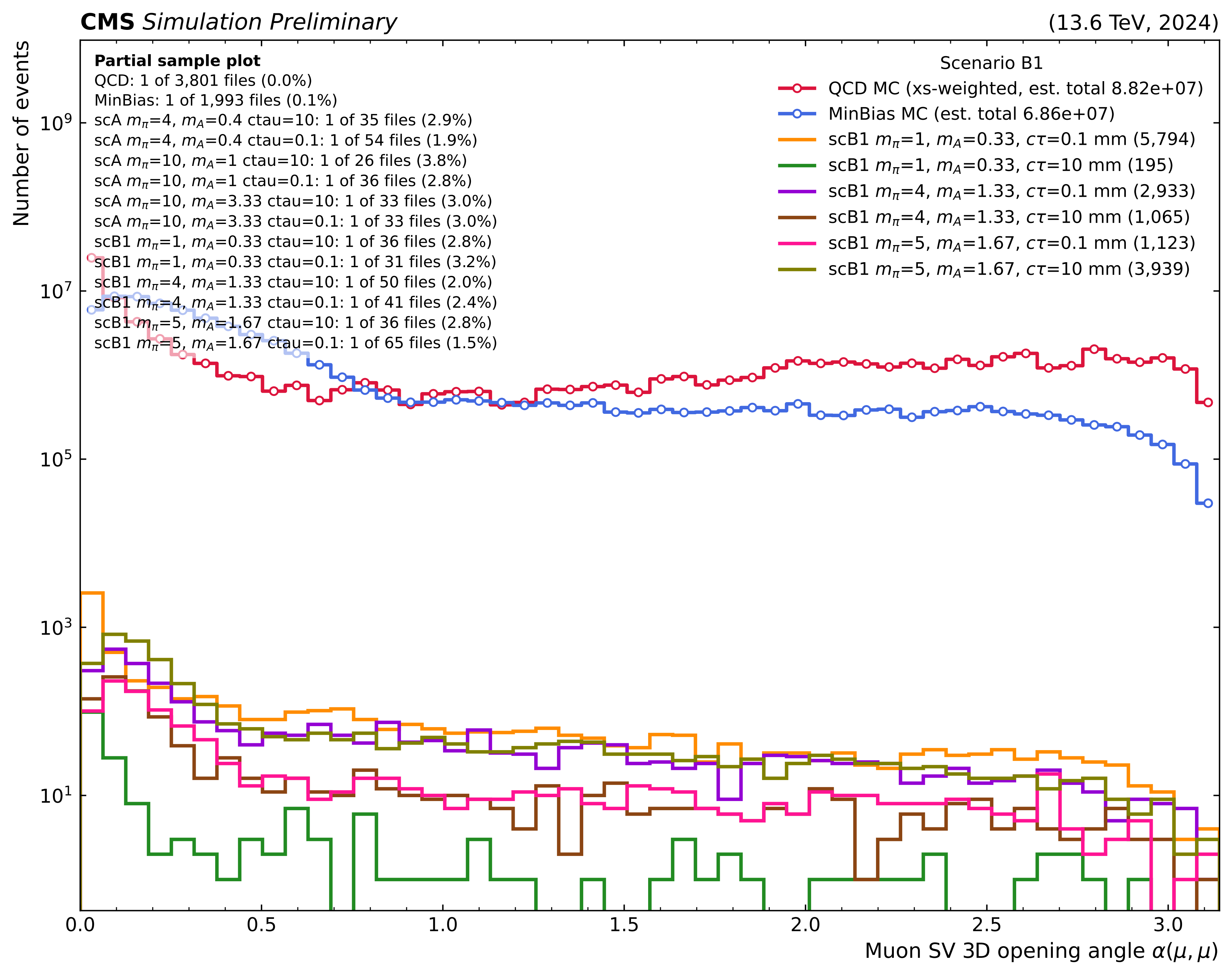
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
Number of events
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$


CMS Simulation Preliminary
(13.6 TeV, 2024)
Number of events
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario A
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm (1,175)
scA m_π=4, m_A=0.4, $c\tau$=10 mm (4,164)
scA m_π=10, m_A=1, $c\tau$=0.1 mm (493)
scA m_π=10, m_A=1, $c\tau$=10 mm (3,564)
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm (4,674)
scA m_π=10, m_A=3.33, $c\tau$=10 mm (2,919)
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$

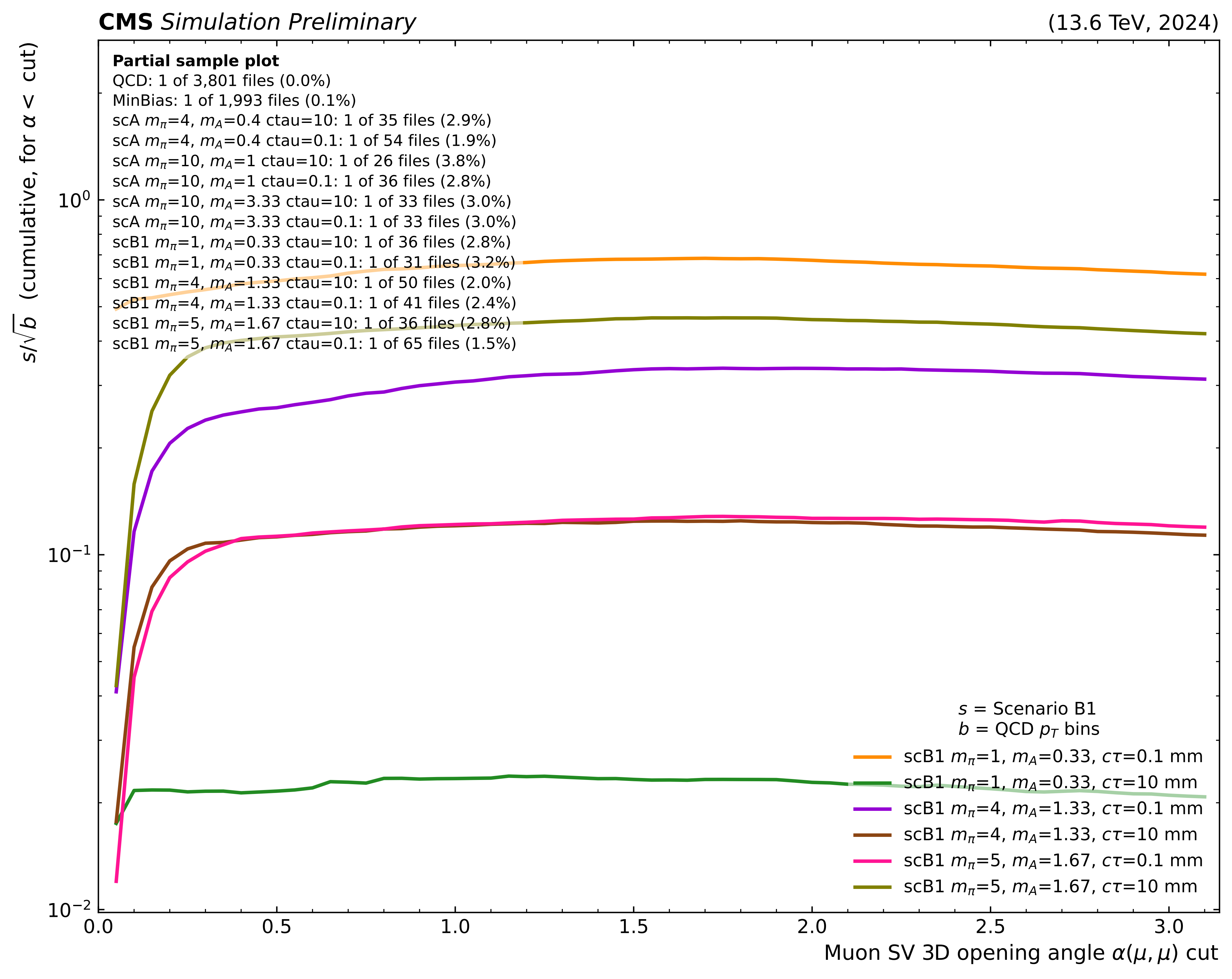


CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $\alpha <$ cut)
10^0
10^{-1}
10^{-2}
s = Scenario A
b = QCD p_T bins
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu, \mu)$ cut

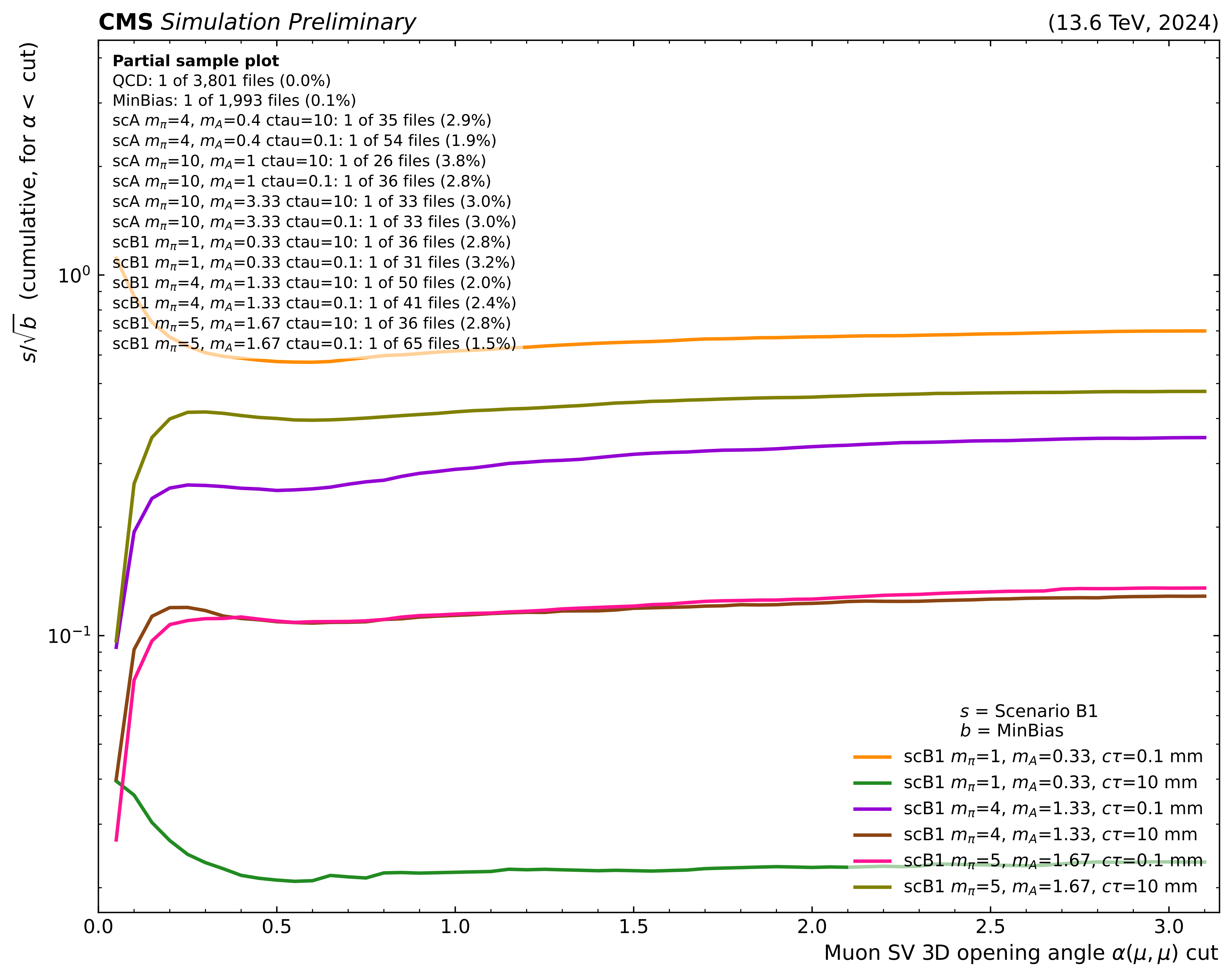

CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $\alpha <$ cut)
10^0
10^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$ cut
s = Scenario A
b = MinBias
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm



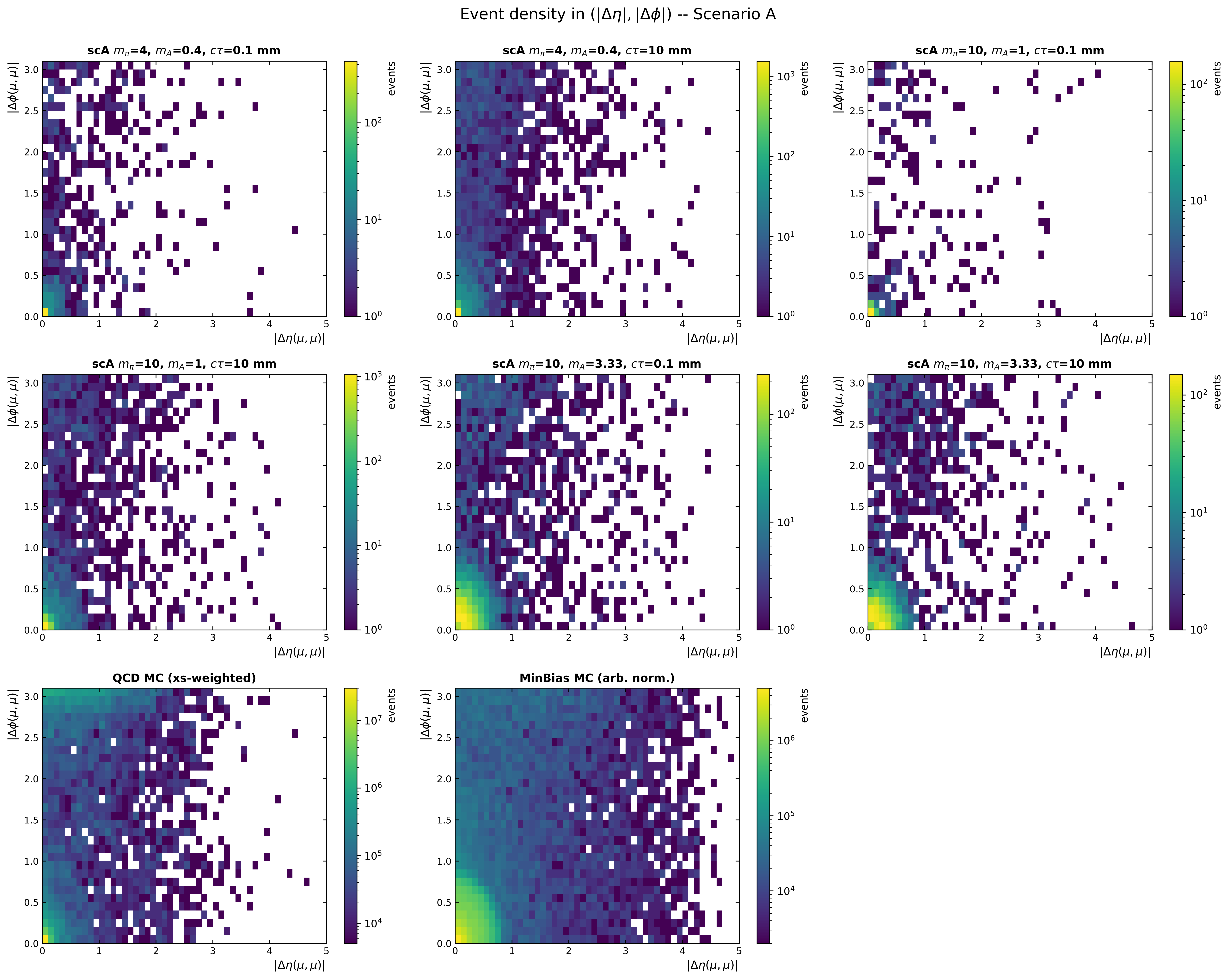
CMS Simulation Preliminary
(13.6 TeV, 2024)
Number of events
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
Scenario B1
QCD MC (xs-weighted, est. total 8.82e+07)
MinBias MC (est. total 6.86e+07)
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm (5,794)
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm (195)
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm (2,933)
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm (1,065)
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm (1,123)
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm (3,939)
10^9
10^7
10^5
10^3
10^1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$



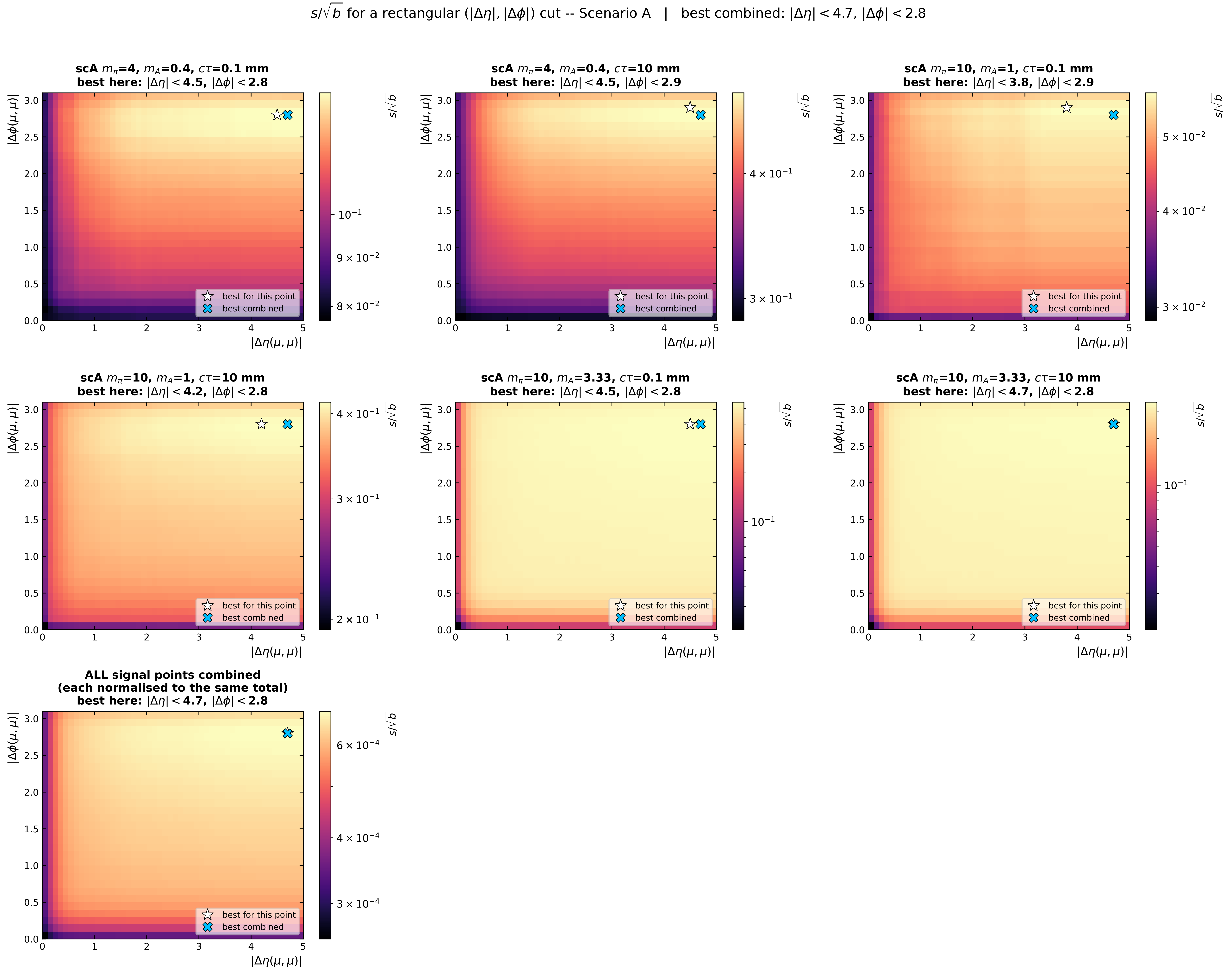
CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $\alpha <$ cut)
10^0
10^{-1}
10^{-2}
s = Scenario B1
b = QCD p_T bins
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$ cut



CMS Simulation Preliminary
(13.6 TeV, 2024)
Partial sample plot
QCD: 1 of 3,801 files (0.0%)
MinBias: 1 of 1,993 files (0.1%)
scA m_π=4, m_A=0.4 ctau=10: 1 of 35 files (2.9%)
scA m_π=4, m_A=0.4 ctau=0.1: 1 of 54 files (1.9%)
scA m_π=10, m_A=1 ctau=10: 1 of 26 files (3.8%)
scA m_π=10, m_A=1 ctau=0.1: 1 of 36 files (2.8%)
scA m_π=10, m_A=3.33 ctau=10: 1 of 33 files (3.0%)
scA m_π=10, m_A=3.33 ctau=0.1: 1 of 33 files (3.0%)
scB1 m_π=1, m_A=0.33 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=1, m_A=0.33 ctau=0.1: 1 of 31 files (3.2%)
scB1 m_π=4, m_A=1.33 ctau=10: 1 of 50 files (2.0%)
scB1 m_π=4, m_A=1.33 ctau=0.1: 1 of 41 files (2.4%)
scB1 m_π=5, m_A=1.67 ctau=10: 1 of 36 files (2.8%)
scB1 m_π=5, m_A=1.67 ctau=0.1: 1 of 65 files (1.5%)
$s/\sqrt{b}$ (cumulative, for $\alpha <$ cut)
10^0
10^{-1}
s = Scenario B1
b = MinBias
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Muon SV 3D opening angle $\alpha(\mu,\mu)$ cut

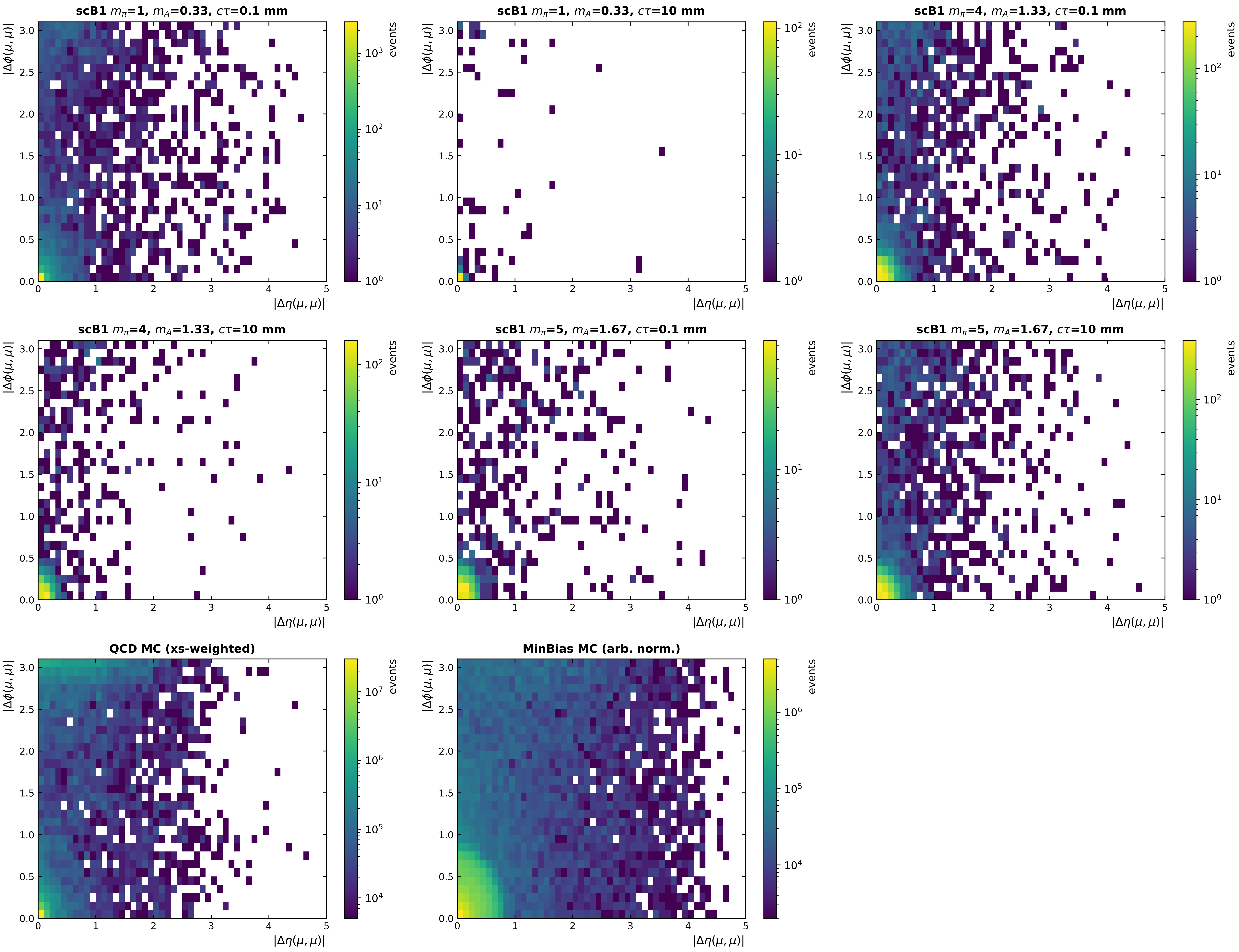


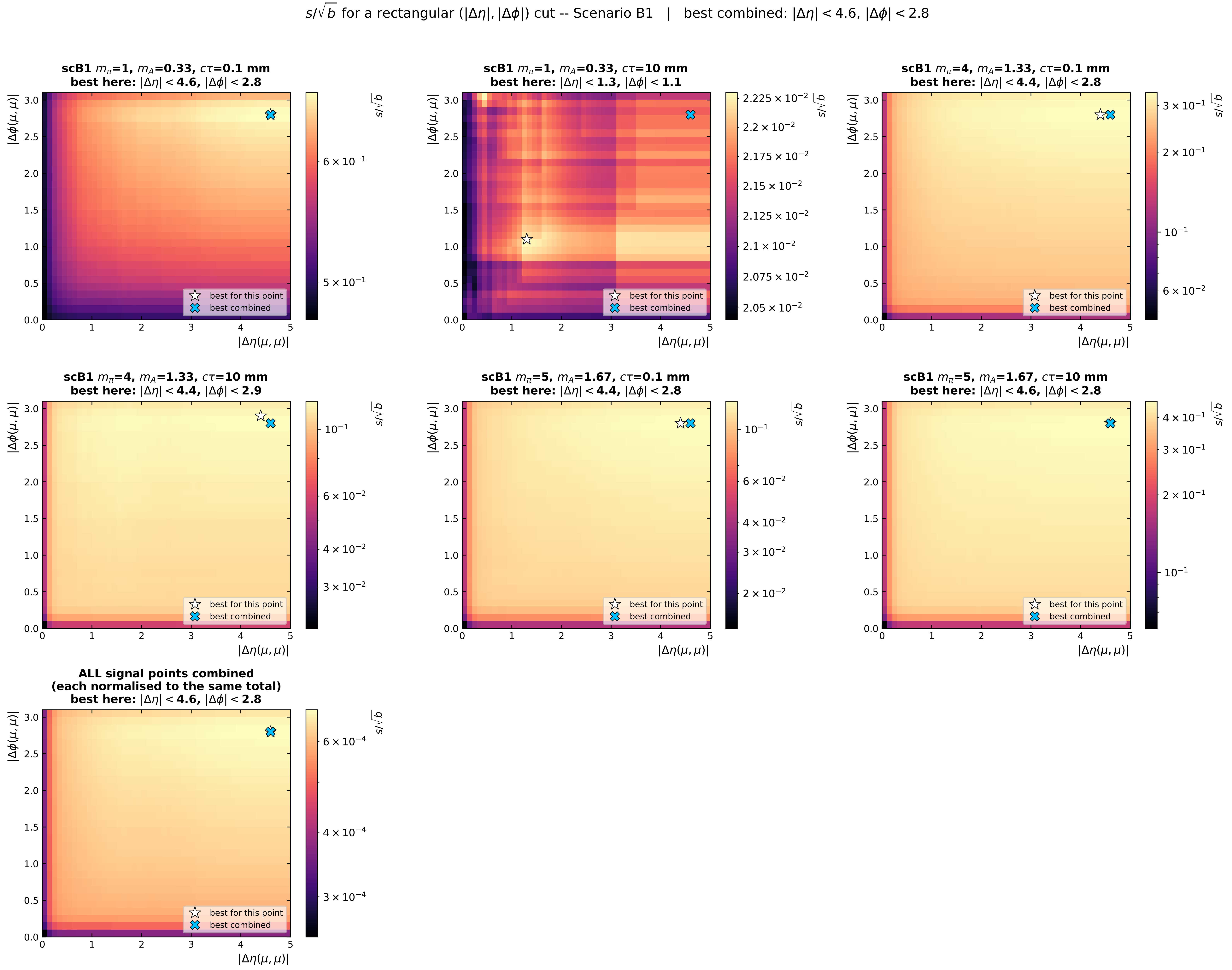
Event density in (|Δη|, |Δϕ|) -- Scenario A
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
scA m_π=4, m_A=0.4, $c\tau$=10 mm
scA m_π=10, m_A=1, $c\tau$=0.1 mm
scA m_π=10, m_A=1, $c\tau$=10 mm
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
scA m_π=10, m_A=3.33, $c\tau$=10 mm
QCD MC (xs-weighted)
MinBias MC (arb. norm.)
$|\Delta\phi(\mu,\mu)|$
$|\Delta\eta(\mu,\mu)|$
events



$s/\sqrt{b}$ for a rectangular ($|\Delta\eta|$, $|\Delta\phi|$) cut -- Scenario A | best combined: $|\Delta\eta| < 4.7$, $|\Delta\phi| < 2.8$
scA m_π=4, m_A=0.4, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 4.5$, $|\Delta\phi| < 2.8$
scA m_π=4, m_A=0.4, $c\tau$=10 mm
best here: $|\Delta\eta| < 4.5$, $|\Delta\phi| < 2.9$
scA m_π=10, m_A=1, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 3.8$, $|\Delta\phi| < 2.9$
scA m_π=10, m_A=1, $c\tau$=10 mm
best here: $|\Delta\eta| < 4.2$, $|\Delta\phi| < 2.8$
scA m_π=10, m_A=3.33, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 4.5$, $|\Delta\phi| < 2.8$
scA m_π=10, m_A=3.33, $c\tau$=10 mm
best here: $|\Delta\eta| < 4.7$, $|\Delta\phi| < 2.8$
ALL signal points combined
(each normalised to the same total)
best here: $|\Delta\eta| < 4.7$, $|\Delta\phi| < 2.8$
$|\Delta\phi(\mu,\mu)|$
$|\Delta\eta(\mu,\mu)|$
$s/\sqrt{b}$
best for this point
best combined

Event density in ($|\Delta\eta|$, $|\Delta\phi|$) -- Scenario B1





$s/\sqrt{b}$ for a rectangular ($|\Delta\eta|$, $|\Delta\phi|$) cut -- Scenario B1 | best combined: $|\Delta\eta| < 4.6$, $|\Delta\phi| < 2.8$
scB1 m_π=1, m_A=0.33, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 4.6$, $|\Delta\phi| < 2.8$
scB1 m_π=1, m_A=0.33, $c\tau$=10 mm
best here: $|\Delta\eta| < 1.3$, $|\Delta\phi| < 1.1$
scB1 m_π=4, m_A=1.33, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 4.4$, $|\Delta\phi| < 2.8$
scB1 m_π=4, m_A=1.33, $c\tau$=10 mm
best here: $|\Delta\eta| < 4.4$, $|\Delta\phi| < 2.9$
scB1 m_π=5, m_A=1.67, $c\tau$=0.1 mm
best here: $|\Delta\eta| < 4.4$, $|\Delta\phi| < 2.8$
scB1 m_π=5, m_A=1.67, $c\tau$=10 mm
best here: $|\Delta\eta| < 4.6$, $|\Delta\phi| < 2.8$
ALL signal points combined
(each normalised to the same total)
best here: $|\Delta\eta| < 4.6$, $|\Delta\phi| < 2.8$
$|\Delta\phi(\mu,\mu)|$
$|\Delta\eta(\mu,\mu)|$
$s/\sqrt{b}$
best for this point
best combined